

Digital Communication System



Unit 1: Introduction of Communication System

- 1.1 Introduction of Electronic communication System
- 1.2 Wave Property and Characteristics.
- 1.3 Electromagnetic Spectrum, Bandwidth and Information Capacity
- 1.4 Signal Analysis
- 1.5 Analog to Digital Conversion and Digital to Analog Conversion
- 1.6 Pulse code Modulation(PCM)
- 1.7 Digital Modulation and Transmission Techniques (ASK, FSK, PSK)

What is Communication?

The act of sharing, imparting or exchanging information, ideas or feelings.



Types of Communication

Verbal



Non Verbal



Written



Visual



What is Electronic Communication?

In Electronics engineering the **communication** is defined as **the process of broadcasting, transmitting, storing or viewing the data and information** using electronic devices and circuits.



Means of Electronic Communication

Telephone



Video Calling



E Mail



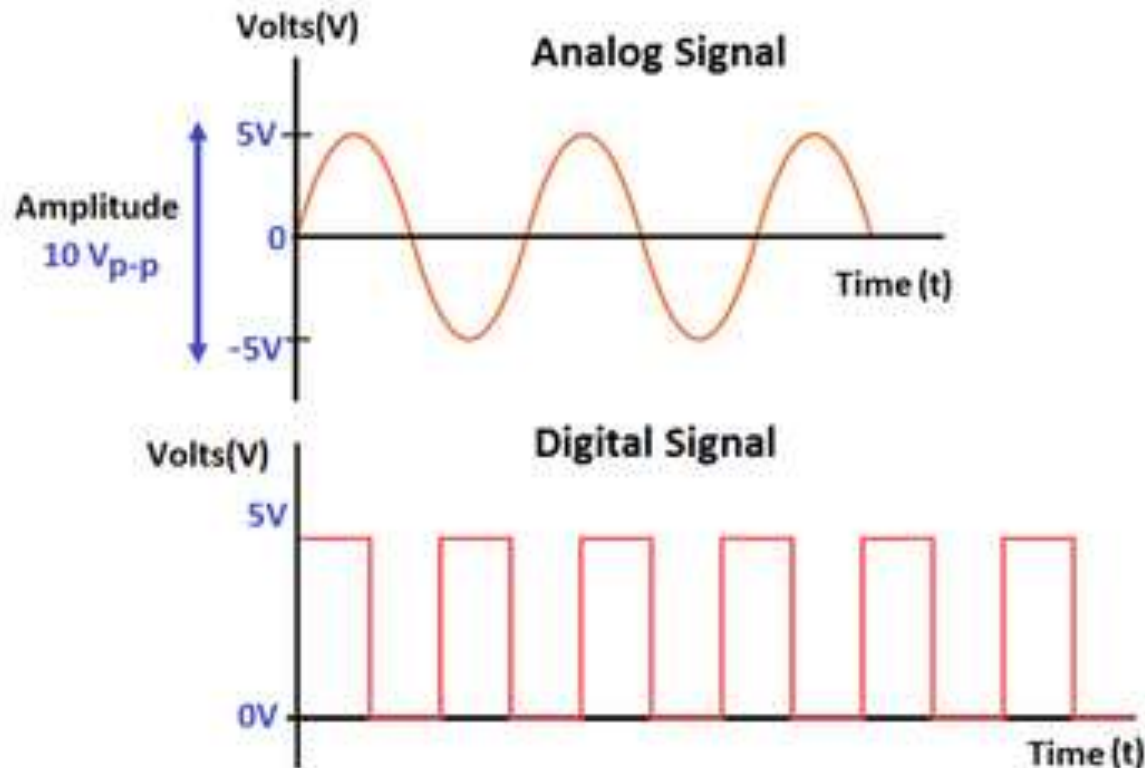
Television



2 Types of Electronic Communication

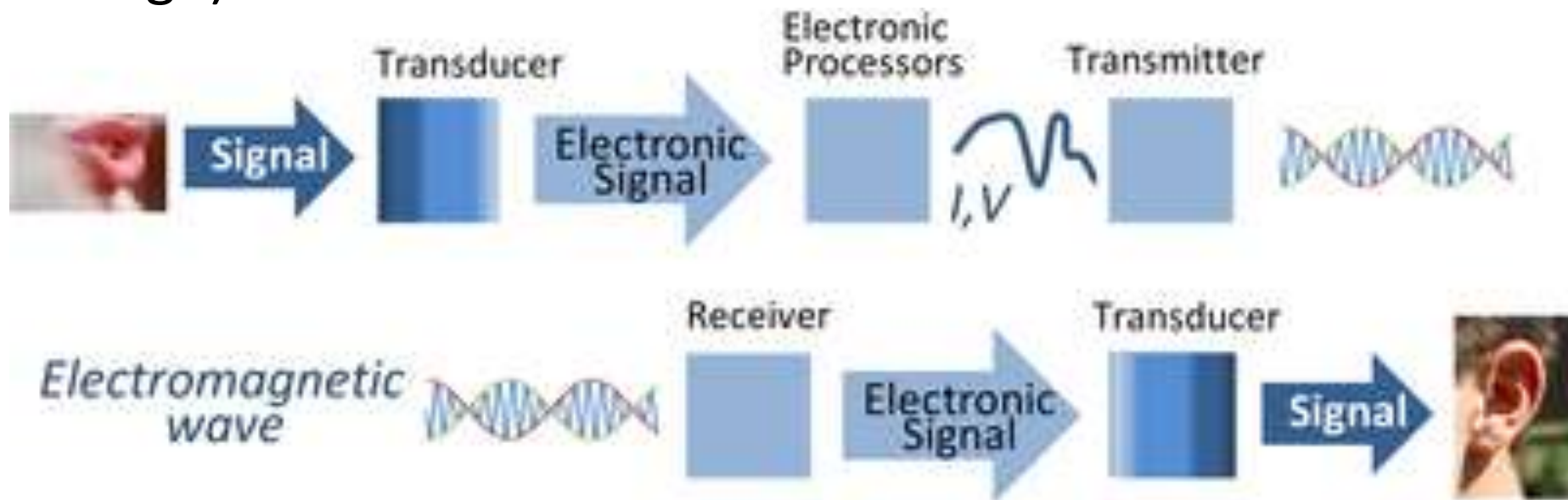
Based on the **nature of signals and devices used for processing the information**, there are two types of Electronic Communication :

- 1) Analog Communication 2) Digital Communication**



Signal in Electronic Communication

Meaning : Information converted into an electrical form suitable for transmission is called a signal. There are two types of signals; Analog and Digital. **Analog signals** are continuous variations of current and voltage whereas **digital signals** are those that have discrete stepwise value (0 = Low, 1 = High)



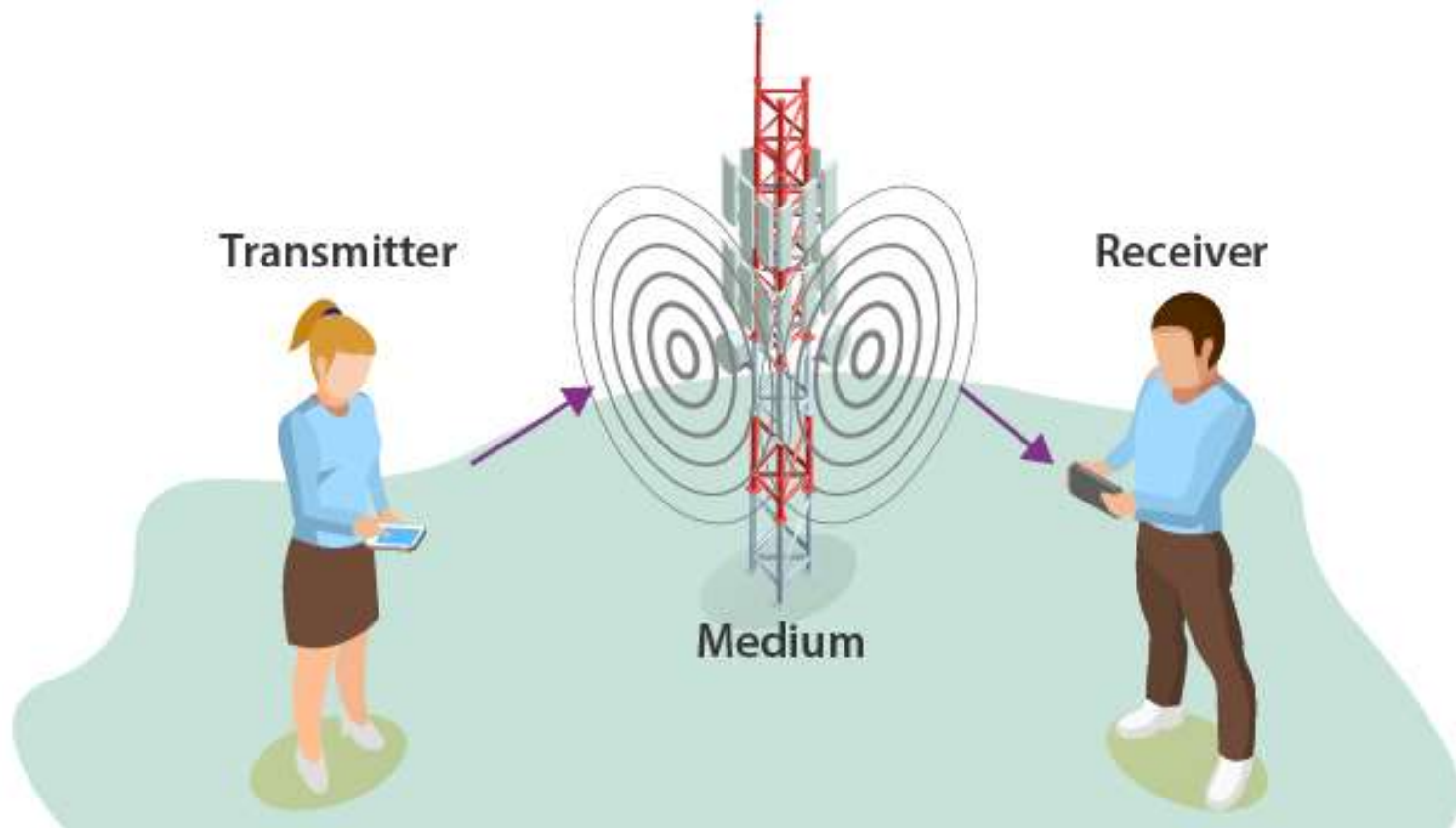
Communication System

There are 3 major components of Communication System:

Transmitter – Sends information

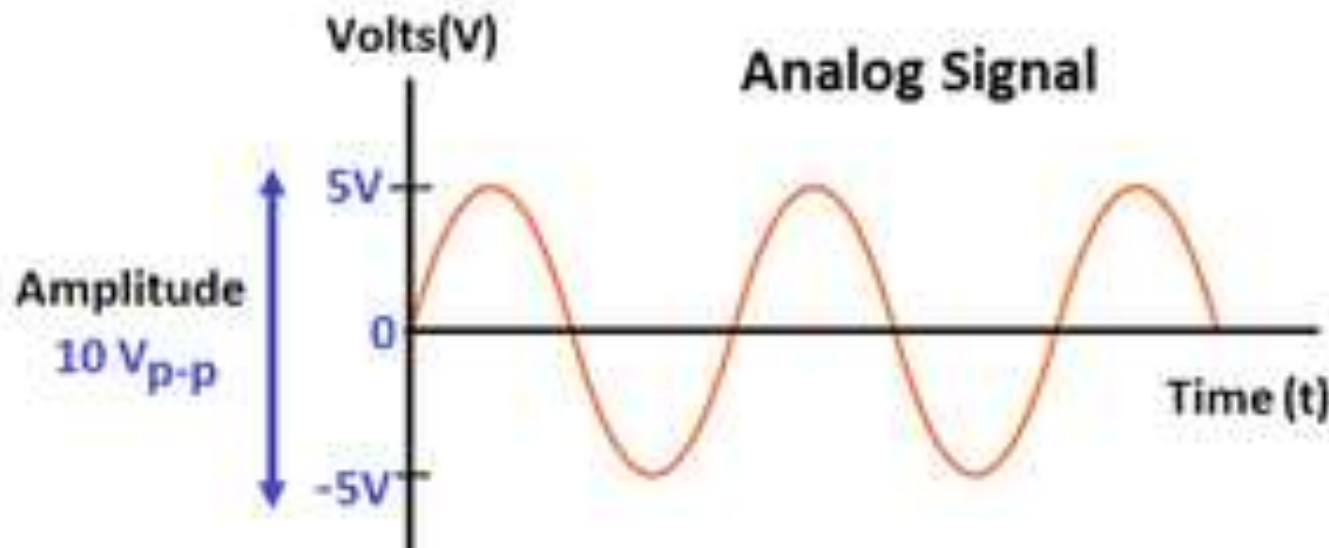
Medium – Channel that carries information

Receiver – Receives information



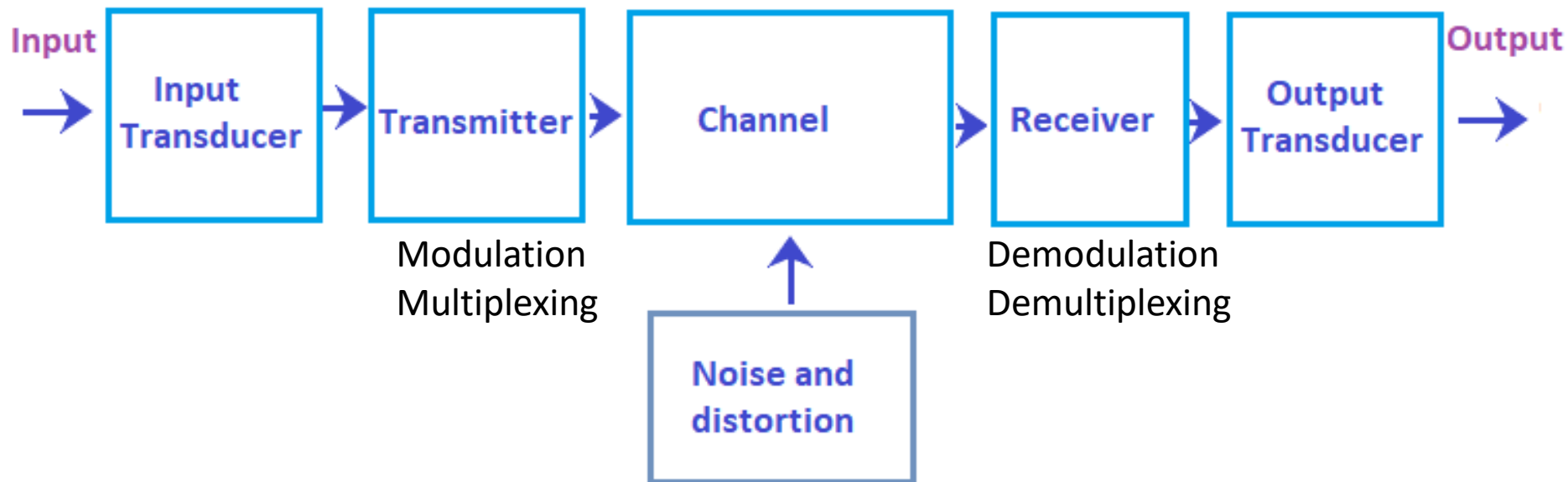
Analog Communication

Meaning : The type of communication in which the data and information is transferred with the help of analog signals in between transmitter and receiver is called the **analog communication**. Therefore, the analog communication uses **continuous time signals** for transmission of information.



Analog Communication

Process: The information in the form of analog signal is input to a **transducer**, which supplies it to the **transmitter**. The **transmitter** broadcast the modulated information on a **communication channel**. At **receiver** end, a **demodulator circuit** is employed for extracting the information from the modulated signal to produce the output message.



Major Components of Communication

Input Transducer : It converts a signal from one form of energy to another. Eg: Microphone, Loudspeakers, Antenna, Sensors, etc.

Transducers are often employed at the boundaries of automation, measurement, and control systems, where electrical signals are converted to and from other physical quantities (energy, force, torque, light, motion, position, etc.).

The process of converting one form of energy to another is known as transduction.

Major Components of Communication

Transmitter : The transmitter converts the electrical signal into a form suitable for transmission for the channel. It performs **modulation** by superimposing the message signal on the **high-frequency** carrier signal. Thus, different channels have different types of the transmitter. If the channel's characteristic varies, the transmitter must adjust itself to maintain the desired range for effective communication.

The original signal is known as the message signal or the baseband signal. Transmitter also performs **multiplexing**, i.e., simultaneous transmission of several signals.

Major Components of Communication

Communication Channel : Communication channel is a **medium to transmit the electrical signal** from transmitter to the receiver. The communication can be **broadcast** or **point to point**. Broadcast refers to a single sender and multiple receivers, such as radio. Point to point communication refers to the communication between a single sender and a single receiver, such as a telephone. The essential parameter for suitable transmission is bandwidth. **The greater the bandwidth, the better will be the transmission.**

The communication channel is further categorized as:

Wired channel

Wireless channel

Major Components of Communication

Interference factors : The interference in the channel is termed as **noise** and **attenuation**.

Attenuation is defined as the loss in the strength of the signal. It is also known as distortion. The attenuation is caused by the passive components in the communication system, such as cables and connectors. It is low in optical fiber as compared to other types of media.

Noise is a serious factor in the communication system. It is defined as any unwanted interference in the signal during the transmission. Noise is categorized as:

Internal Noise – Inside communication channel

External Noise – Outside communication channel

Major Components of Communication

Receiver: The receiver receives information from the channel. It extracts the necessary information from the signal required by the output transducer. The receiver performs the opposite of modulation and multiplexing, i.e., **demodulation** and **demultiplexing**. It also **amplifies** and **removes noise** from the signal.

Major Components of Communication

Output Transducer: The output transducer works reversely as that of the input transducer. It converts the electrical energy into the original signal. Examples of output transducers are **loudspeakers, motors, LEDs**, etc.

Both the input and output transducers are important because they convert the signal suitable for transmission and increase the signal's speed.

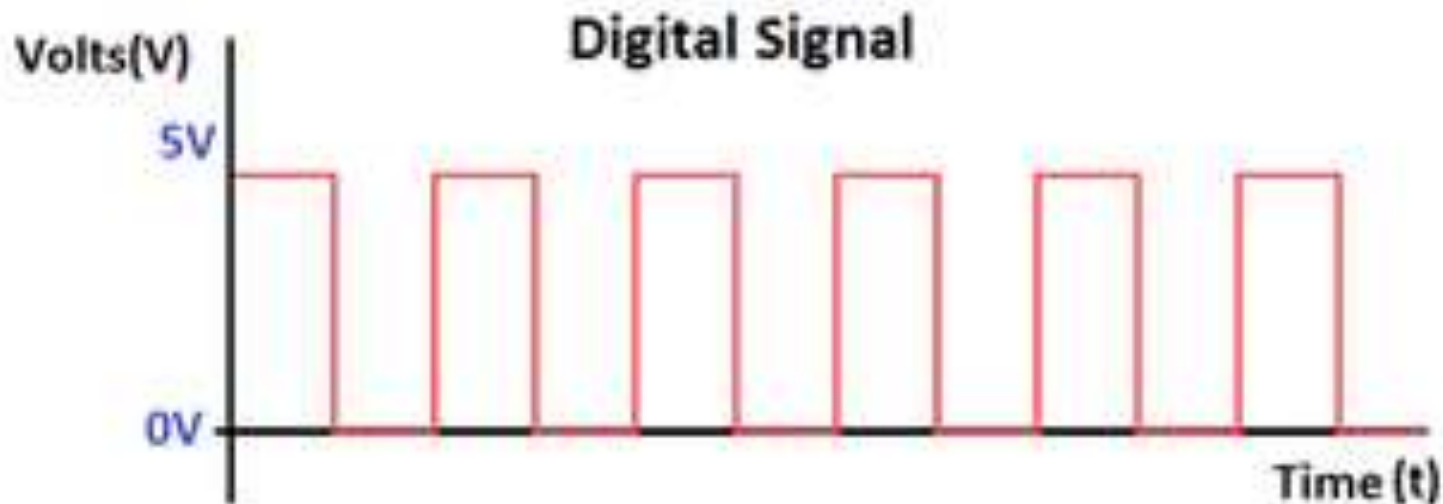
The **loudspeakers** convert the electrical energy to sound.

The **motors** convert the electrical energy into motion.

The **LEDs** (Light Emitting Diodes) convert the electrical energy to light energy.

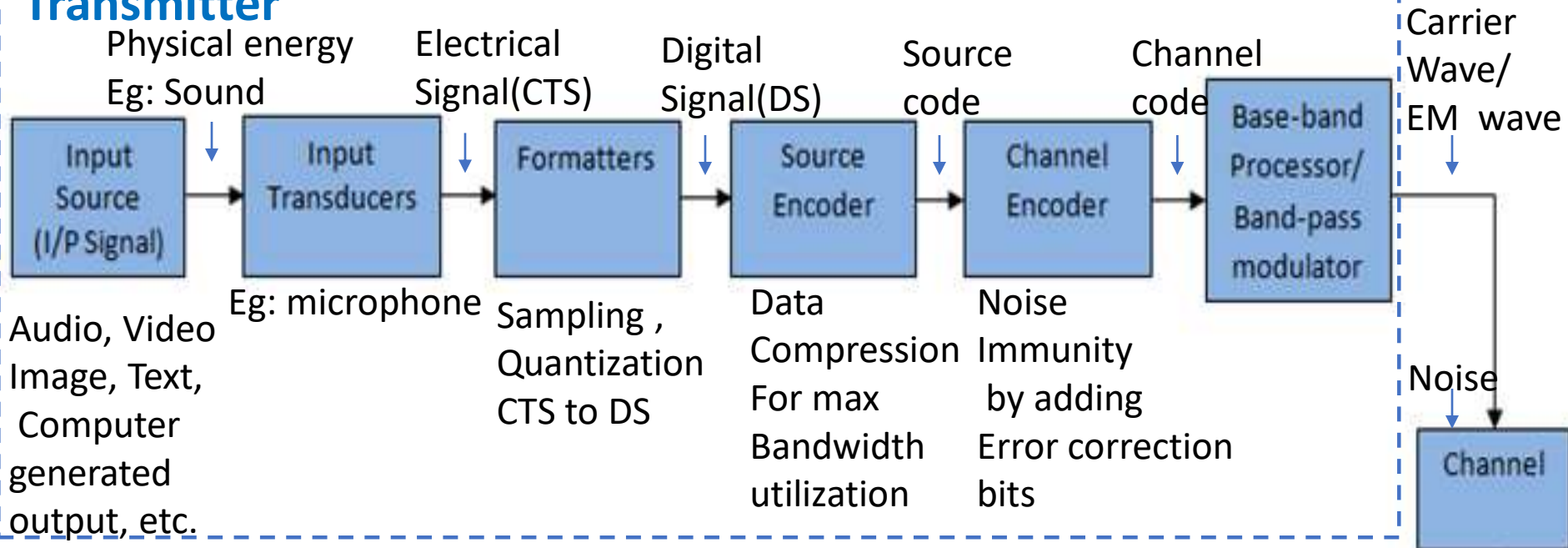
Digital Communication

Meaning : The communication in which the information is transferred by using digital signals in between transmitter and receiver is known as **digital communication**. Hence, in case of digital communication, **discrete time signals** are used for carrying the information from one point to another.



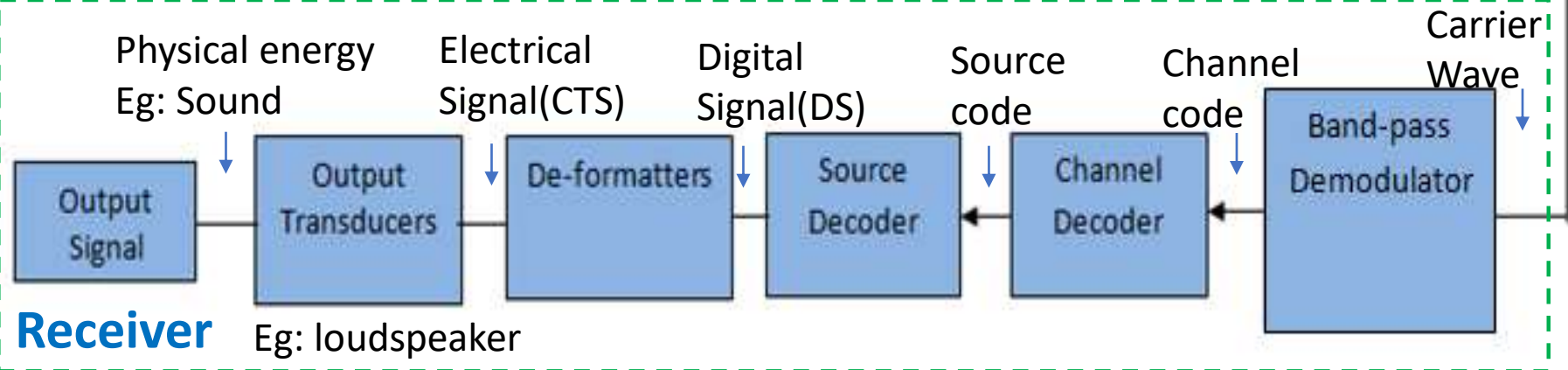
Digital Communication Block Diag.

Transmitter



Receiver

Eg: loudspeaker



Major Components of Communication

Input Source: It refers to the input signal applied to the digital communication system which is generally analog in nature-like voice or video signal. These signals are non-electrical quantities and hence can not be processed directly in a digital communication system.

Input Transducer: A transducer is a device which converts one form of energy into another form. Since the information from the input source are non-electrical quantity, the input transducer convert this non-electrical quantity into electrical quantity.

Major Components of Communication

Formatters: It is used to convert the analog signal into a digital signal. So, the received signal from transducer is converted to a digital signal with the help of formatter. If input signal is digital then no need of formatter.

Source Encoder: It **compresses** the data to the reduced number of bits from the original bits. It helps in effective bandwidth utilization and also **removes unnecessary bits**. It means that the compressed data is in the form of binary digits. The output data is further passed to the channel encoder.

Major Components of Communication

Channel Encoder: It does the coding for error correction. The information in the signal may get altered due to the noise during the transmissions. Hence, it adds some redundant bits to the binary data that helps in correcting the data on the receiver side.

It enhances the transmission quality of the signal and the channel.

Major Components of Communication

Base band Processor/Digital Modulator: It modulates the signal by varying the transmitted signal's frequency, amplitude, and phase. The original signal is converted to high frequency carrier signal.

Channel: It is the media via which the signal is transmitted from the transmitting end to the receiving end. It is the part of model at which maximum noise is added to the signal.

Digital Demodulator: The signal is demodulated and the source signal is recovered from the carrier signal. A device that performs both modulation and demodulation is called a modem -- a name created by combining the first letters of MOdulator and DEModulator.

Major Components of Communication

Channel Decoder: It removes the parity bits from the binary data. It does not affect the signal quality and the information and transmits the data securely. The output of the channel decoder is a **pure digital signal with no interference or noise**.

Source Decoder : The source decoder works oppositely as that of the source encoder. It converts the binary data back to the waveforms.

Deformatters: The deformatter converts the digital data into analog signal.

Major Components of Communication

Output Transducer: The output transducer works in the opposite way as that of the input transducer. It converts the electrical energy back into its original physical form. It makes the information suitable for the user at the output to capture. Eg: Loudspeakers.

Output signal : The output which is produced after the whole process. Eg: Sound.

Analog Vs. Digital Communication

Analog	Digital
Uses Continuous Time Signals	Uses Discrete Time Signals
Signal representation is in sinusoidal wave form	Signal representation is in square waveform
Major components: The main components used for analog communication are: transmitter, transducers, modulator, channel, demodulator and receiver.	The components of digital communication are: transmitter, encoder, modulator, channel, demodulator, decoder and receiver.
Needs low bandwidth	Needs high bandwidth

Analog Vs. Digital Communication

Analog	Digital
Less immunity to noise	High immunity to noise
Not possible to separate noise signal from message signal	Noise can be easily separated from message signal
Probability of error is high	Probability of error is low
Hardware used is more complicated and less flexible	Hardware used is less complicated and more flexible
Information transmission cost is less	Information transmission cost is high
Accuracy is less	Accuracy is more

Wave Property and Characteristics

What is a Wave?

- A wave is a disturbance that carries energy from one place to another.

- It does not move the matter.

Some waves do not need matter (called a “**medium**”) to be able to move (for example, through space).

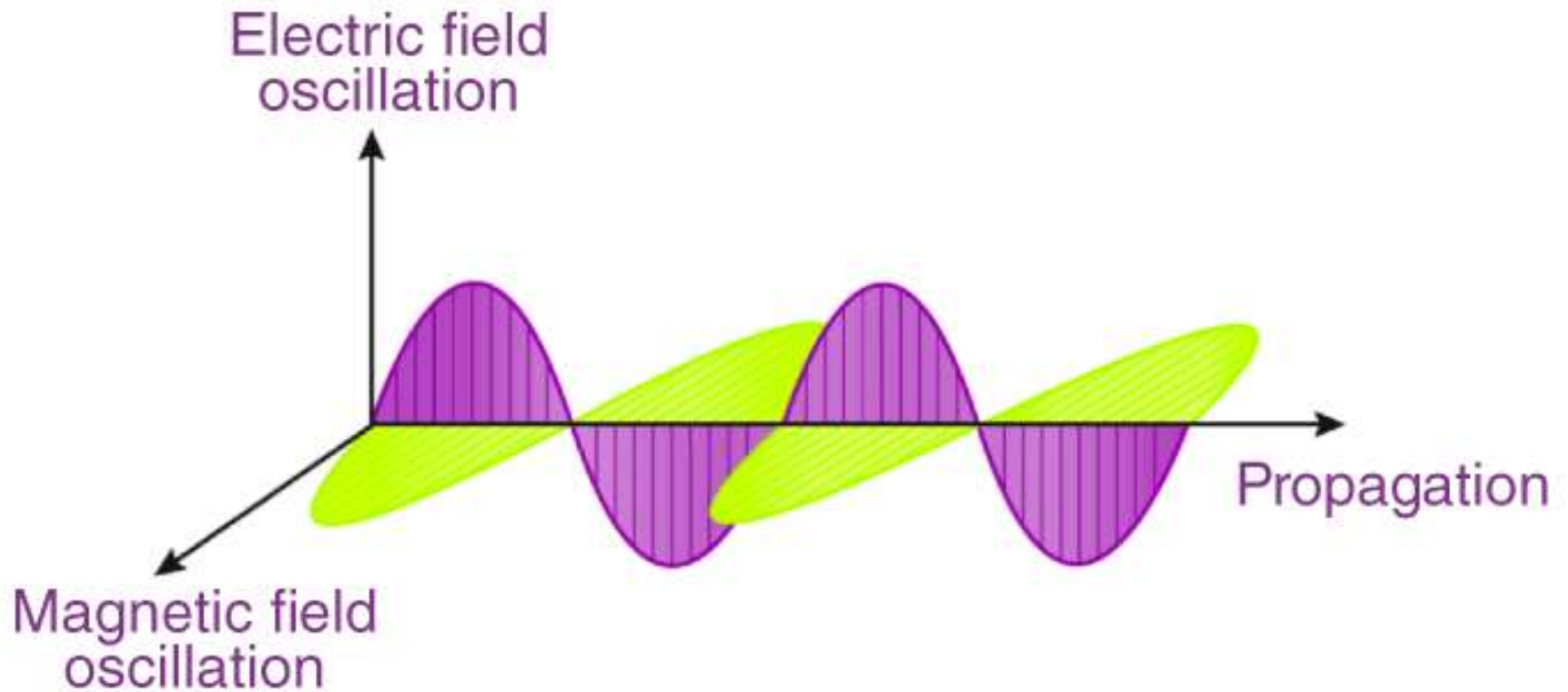
These are called **electromagnetic waves** (or EM waves).

Some waves **MUST** have a medium in order to move. These are called **mechanical waves**.

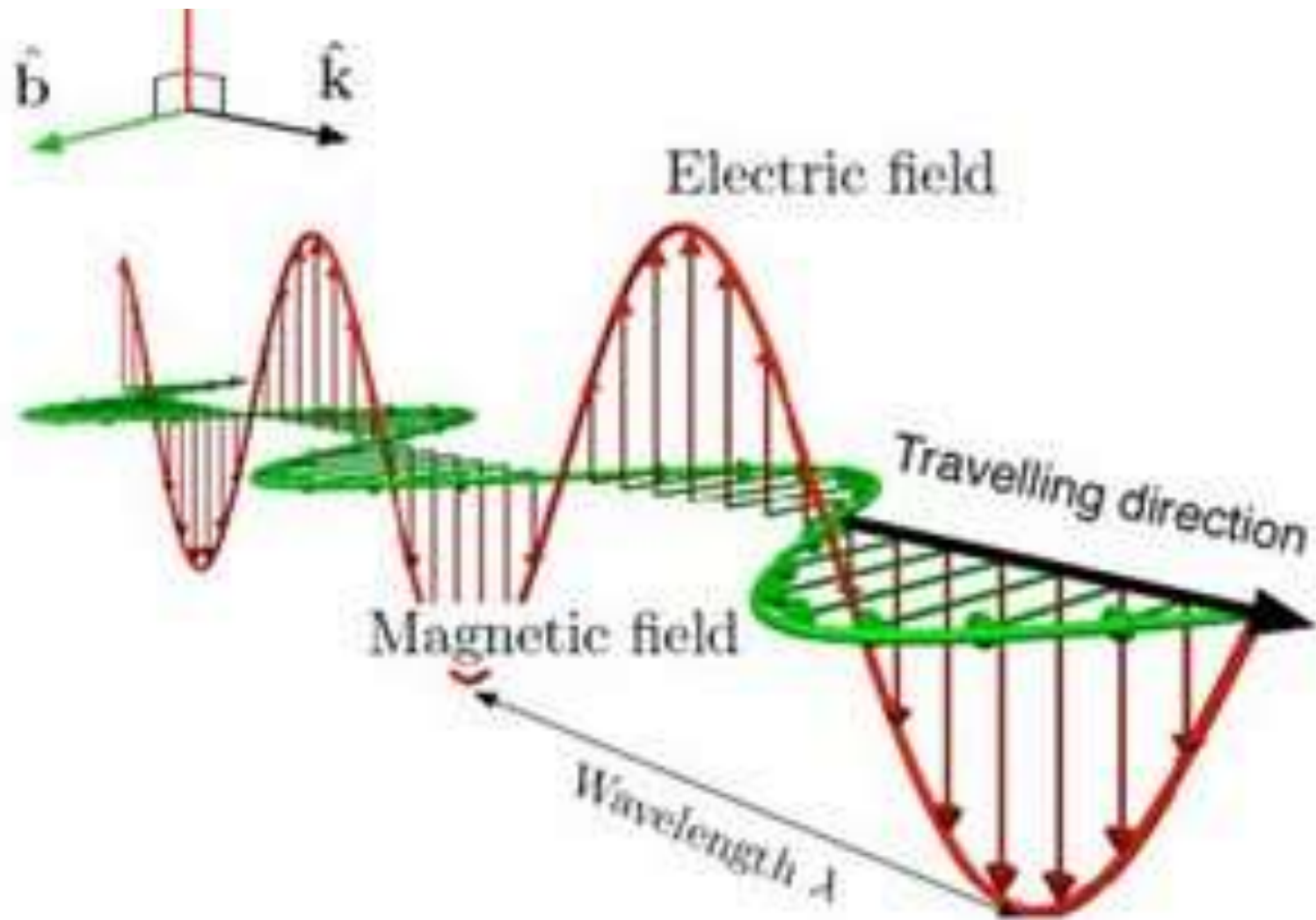
What is an EM Wave?

- Electromagnetic waves or EM waves are waves that are created as a result of **vibrations between an electric field and a magnetic field**.
- The electric field and magnetic field of an electromagnetic wave are perpendicular (at right angles) to each other. They are also perpendicular to the direction of the EM wave.
- An electromagnetic wave can travel through anything - be it air, a solid material or vacuum. **It does not need a medium to propagate** or travel from one place to another.

ELECTROMAGNETIC WAVES



EM Wave



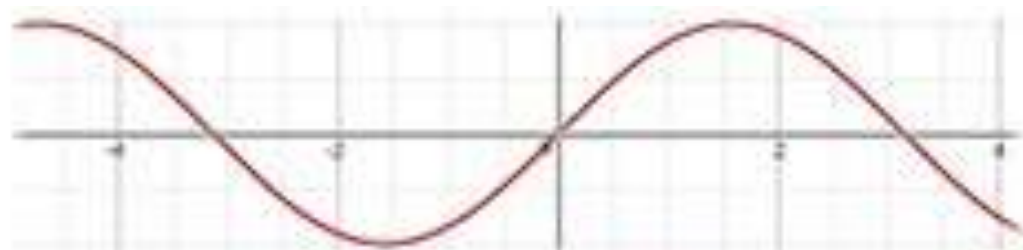
Properties of a Wave

- Amplitude
- Wavelength
- Frequency
- Speed

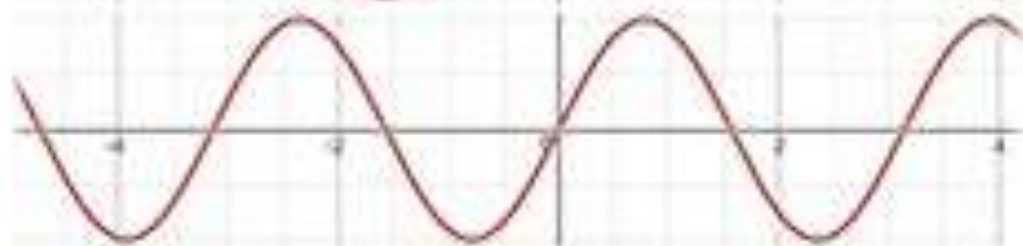
Properties of a Wave

- **Frequency** - The number of cycles of a repetitive wave that occur in a given period of time.
- Frequency is measured in **cycles per second (cps)**.
 - 10 waves going past in 1 second = 10 Hz
 - 1,000 waves go past in 1 second = 1,000 Hz
 - 1 million waves going past = 1 million Hz
- The unit of frequency is the **hertz (Hz)**.
- **The higher the frequency, the more energy in the wave.**

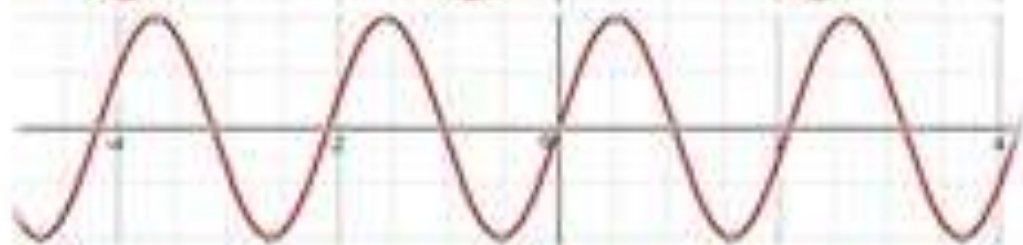
$$\sin(x)$$



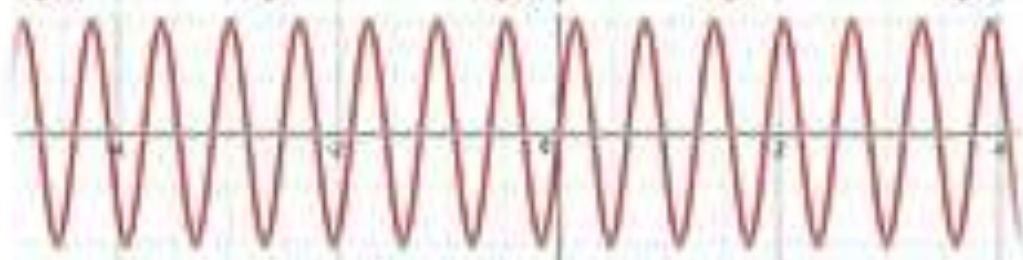
$$\sin(2x)$$



$$\sin(3x)$$



$$\sin(10x)$$



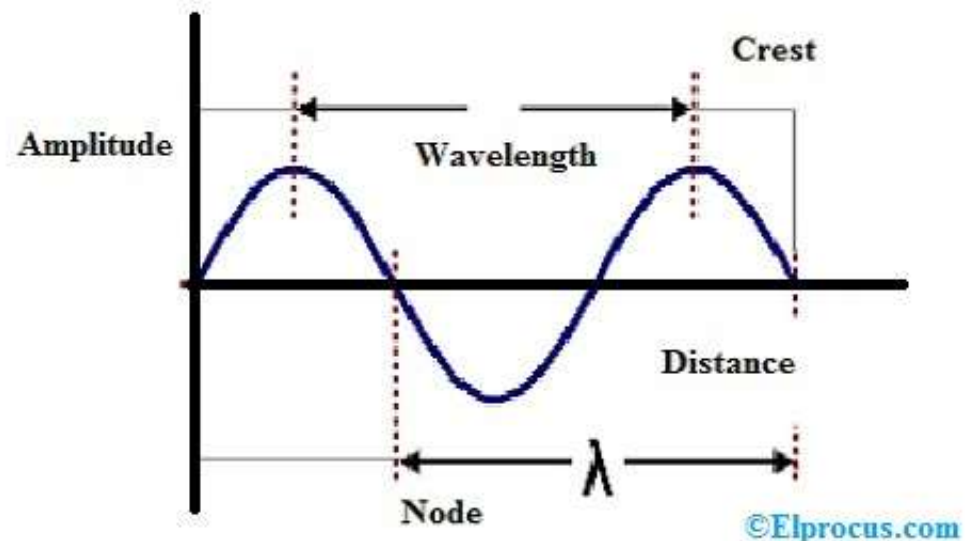
lower
frequency



higher
frequency

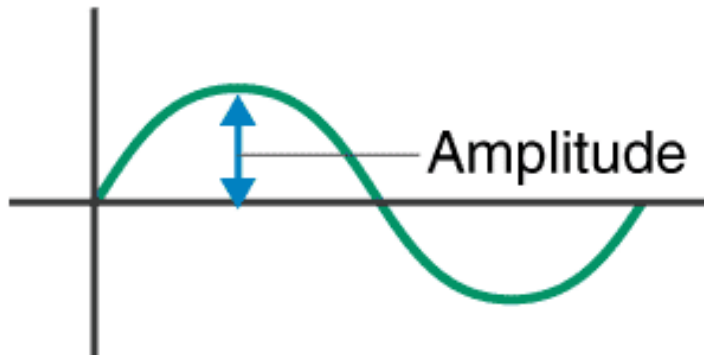
Properties of a Wave

- **Wavelength** - The distance between one point on a wave and the exact same place on the next wave.
- It is also the distance traveled by an electromagnetic wave during the time of one cycle.
- The wavelength of a signal is represented by the Greek letter **lambda** (λ).
- Measured in **metres**.

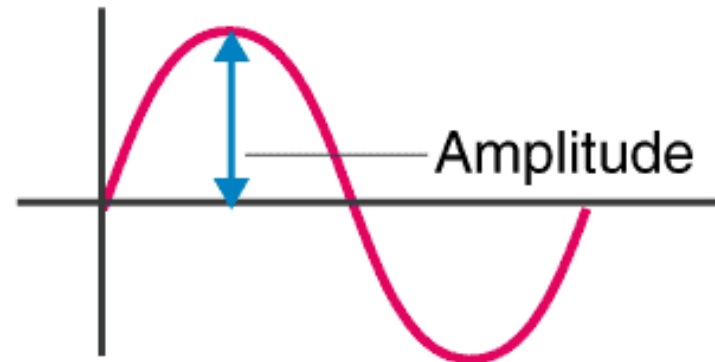


Properties of a Wave

- **Amplitude** - How far the medium moves from rest position (where it is when not moving).
- Remember that for transverse waves, the highest point is the **crest**, and the lowest point is the **trough**.



Quieter

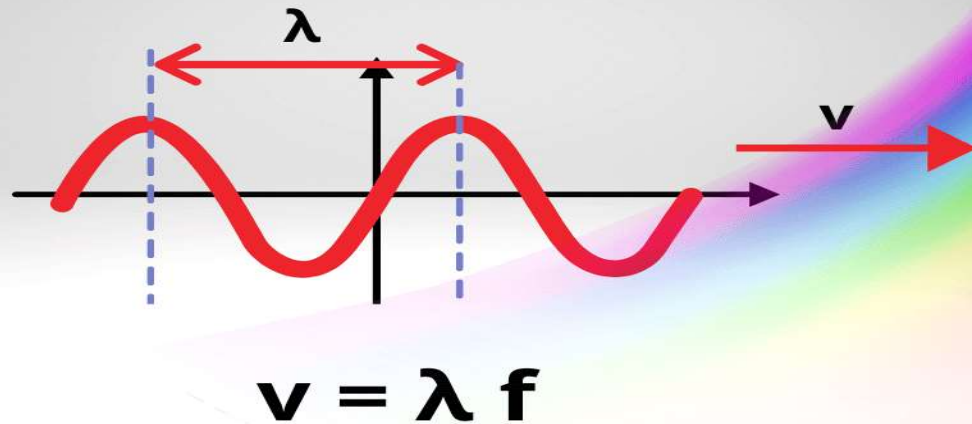


Louder

Properties of Wave

- **Speed of a Wave** - Electromagnetic waves propagate through free space at the velocity of light i.e. $3 \times 10^8 \text{m/s}$ (or 186,000 miles per second).
- These **spread uniformly** in all directions from a point source.

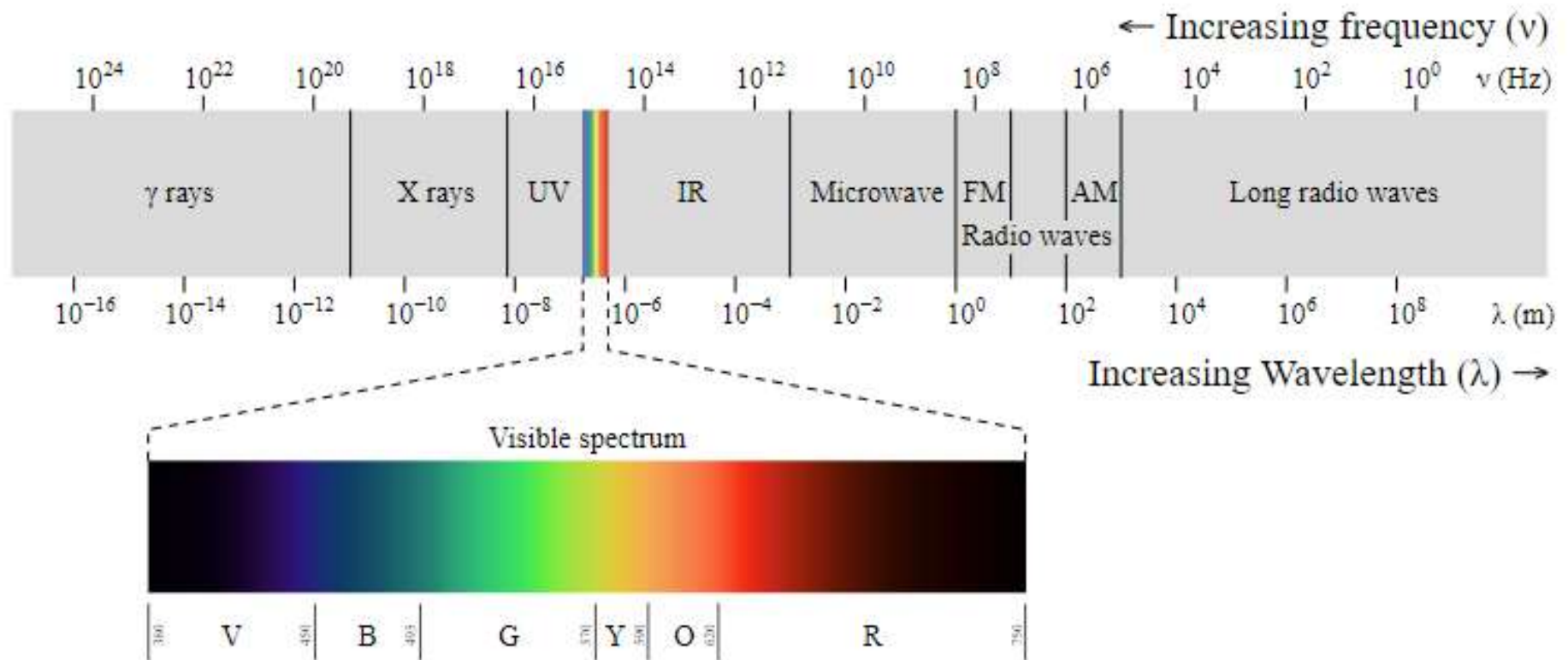
Wavelength to Frequency



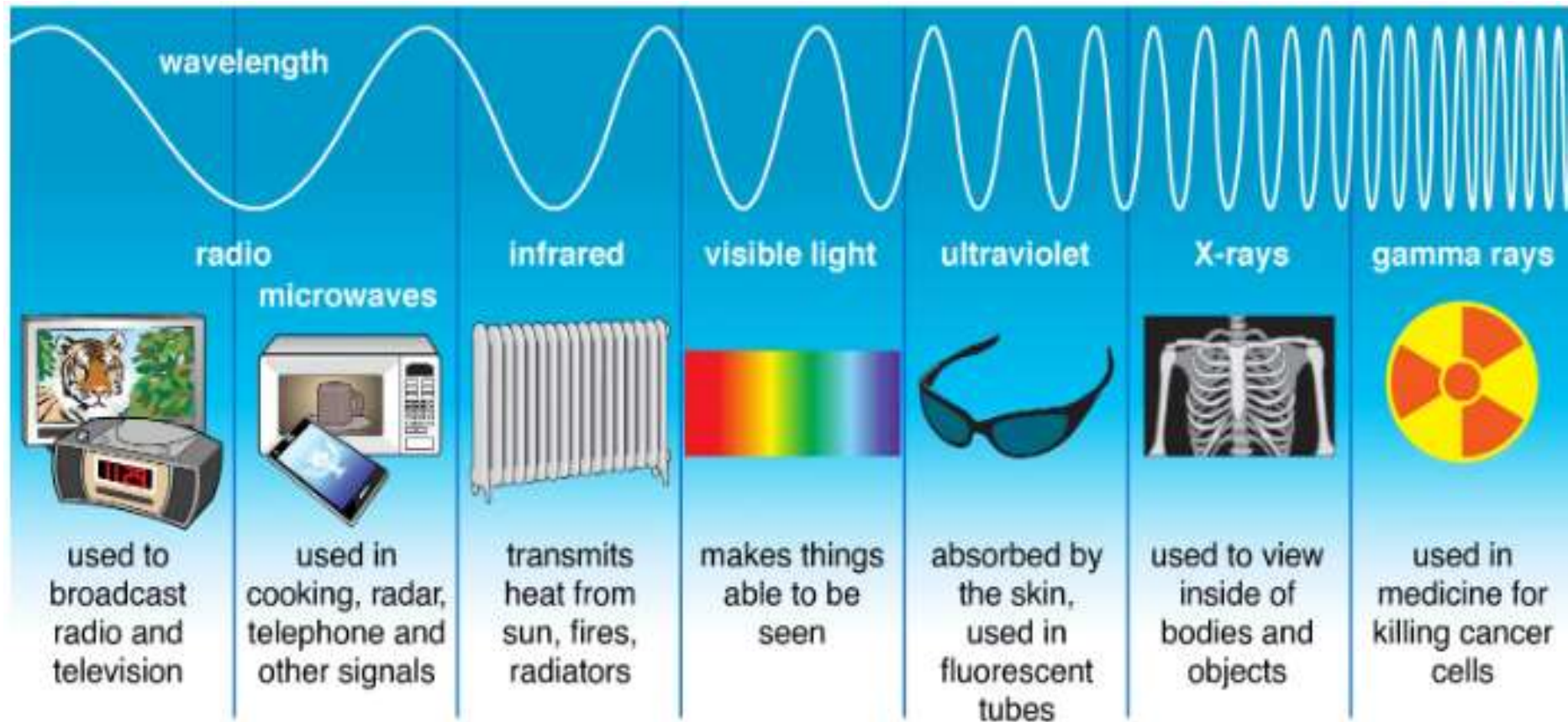
v = wave speed (c for light in a vacuum)
 λ = wavelength

Electromagnetic Spectrum

Meaning : The range of electromagnetic signals encompassing all frequencies is referred to as the **electromagnetic spectrum**.

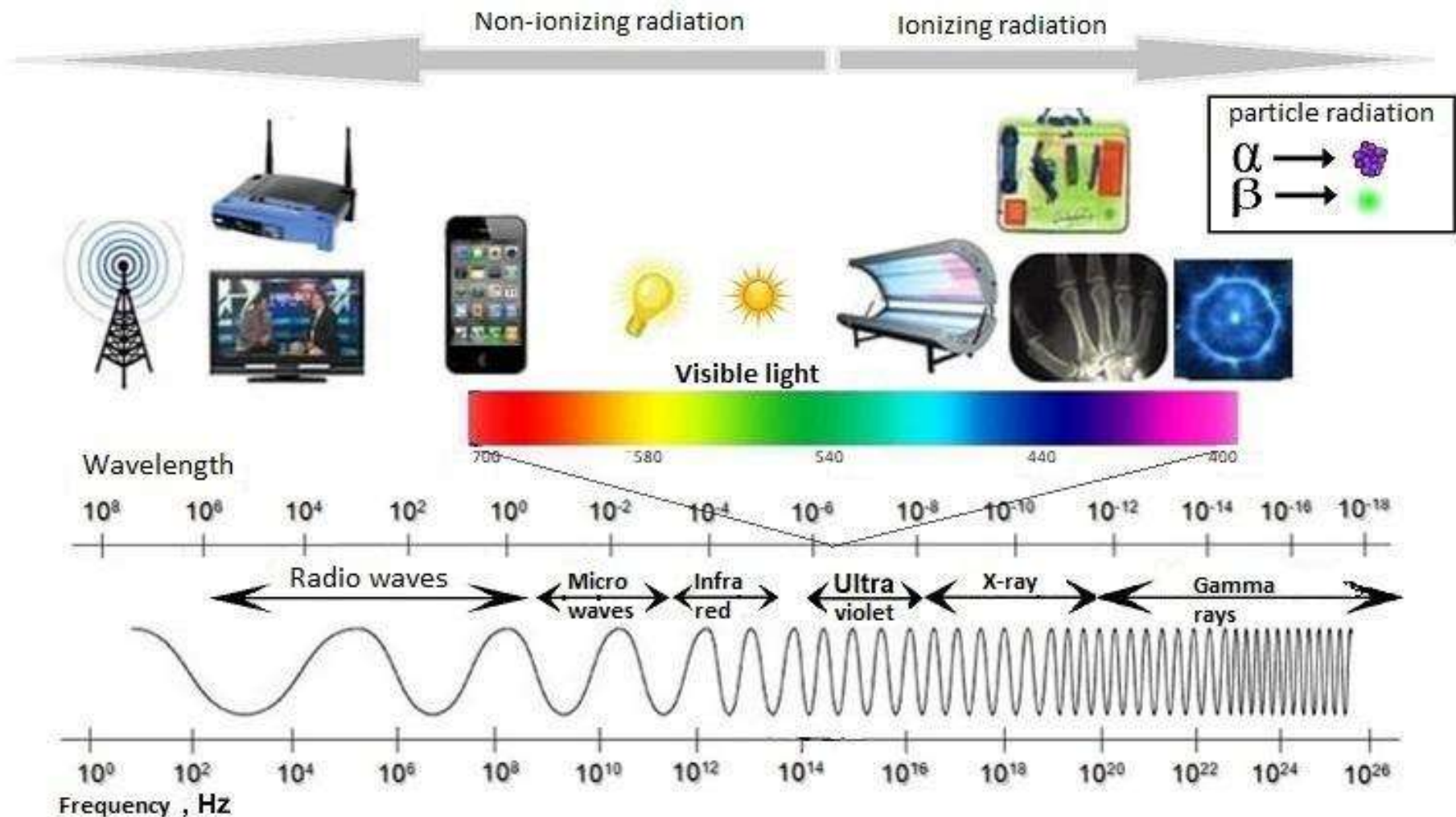


Electromagnetic Spectrum



Electromagnetic Spectrum

The electromagnetic spectrum



Electromagnetic Spectrum

Frequency is inversely proportional to Wavelength

Higher the frequency -> lower Wavelength

Lower the frequency -> higher Wavelength

$$\text{Wavelength} = \frac{\text{Speed of Light}}{\text{Frequency}}$$

$$\lambda = \frac{v}{f}$$

Bandwidth

- **Bandwidth** : The amount of data that can be transferred through a communication medium in a unit time.
- The bandwidth of **digital signal is measured in Bits per second** or Bytes per second.
- The bandwidth of **analog signal is measured in cycles/second** or Hertz.

Baseband

- **Baseband** : It is a communication technique in which the digital signals(original signals) are placed on the transmission line without change in modulation(without any carrier waves).
- It means digital signals are directly transmitted over transmission line.
- It transmits only one signal at a time.
- Digital signals are commonly called Baseband Signal.

Broadband

- **Broadband** : It is a technique to transmit large amount of data such as voice and video over long distances.
- It can send data by modulating each signal onto a different frequency.

Data Transmission

- **Two types:**
 - **Baseband Transmission**
 - **Broadband Transmission**

Data Transmission

1. Baseband Transmission

- Binary data generated by keyboard, data sent to peripheral devices like printers or to other computers in LAN. Low power signal but very less distance.
- The digital signal used in baseband transmission occupies the entire bandwidth of the network media to transmit a single data.
- Used for short distance communication.

Data Transmission

2. Broadband Transmission

- Carrier waves are used to transmit original signals.
- Modulation and Multiplexing is done here.
- Used for long distance communication.
- Eg. Data over Internet.

Baseband Vs. Broadband Transmission

Baseband	Broadband
Single Signal use full bandwidth	Multiple signals use media bandwidth
Signal travels a short distance	Signal travels a long distance
Setup and maintenance cost is low	High

Internet Speed and Bandwidth

Bandwidth (information capacity) – How much data can flow at a time.

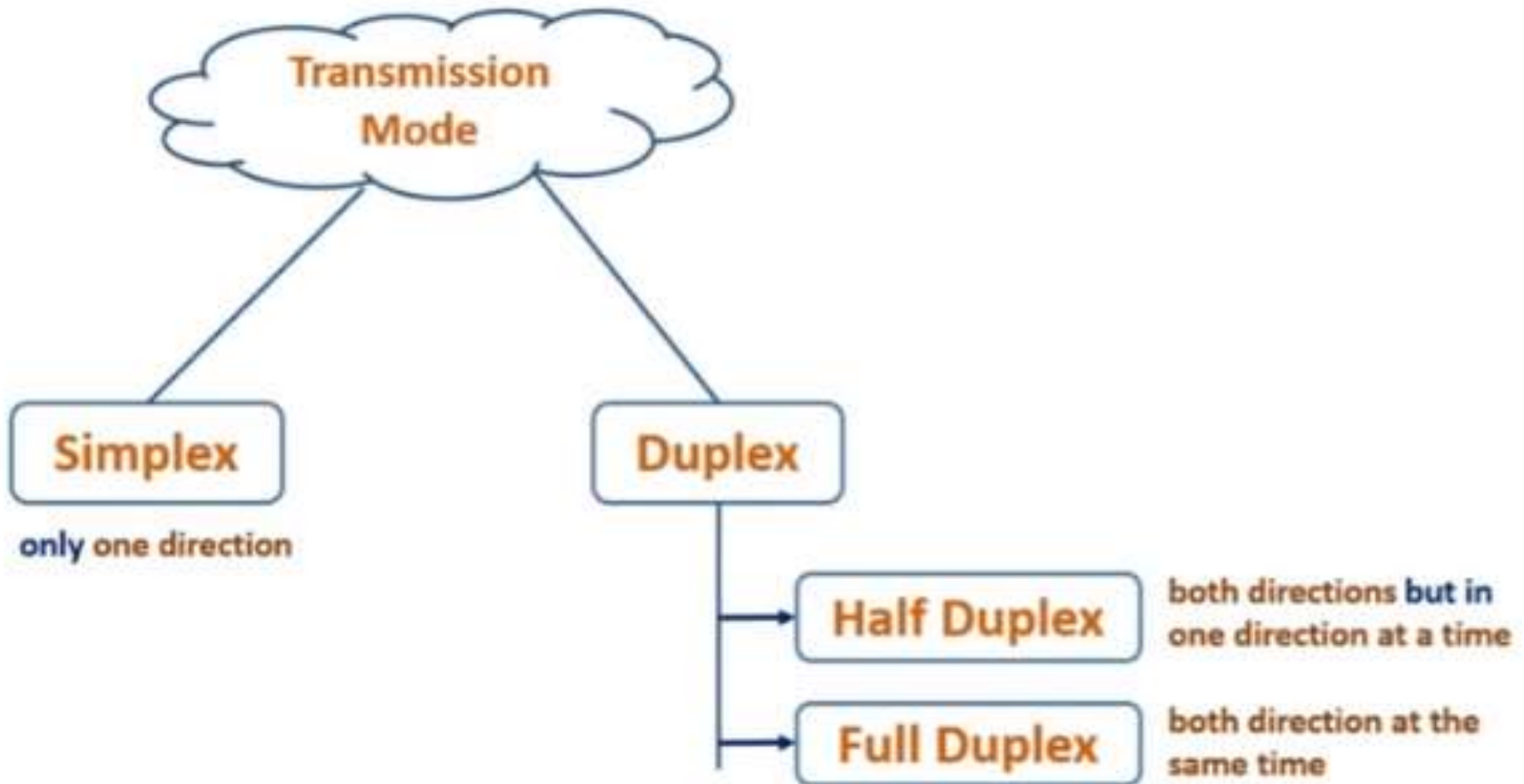
Speed means - How fast the information flows in the channel.

Eg: Web Browsing – You need instant answer. Here bandwidth is not important but speed of information is important.

Netflix, Youtube – You need more data rate transfer for buffering videos. You need here more bandwidth.

Data Transmission Mode

Data Transmission Mode: It defines the **direction of the flow of data** between two communication devices.



Data Transmission Mode

Simplex

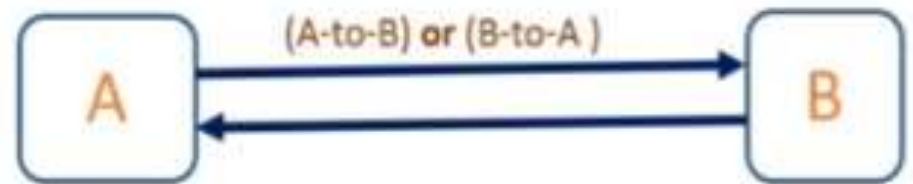


- Data can flow in only **one direction**
- i.e. communication is **unidirectional**.
- In this mode, a sender can only send data but can't receive it similarly , a receiver can only receive data but can't send it.

Eg: Keyboard or mouse inputs to CPU, CPU output to monitor or printer, Scanner to Computer, TV and Radio Signals.

Data Transmission Mode

Half Duplex:

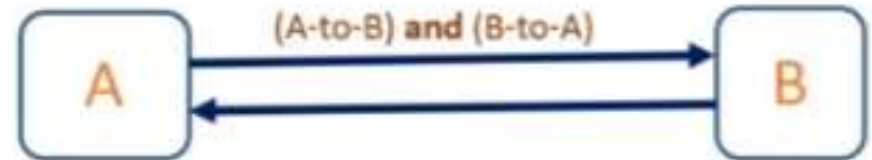


- Data can flow in **both directions** but in **one direction at a time**.
- When one device is sending the other can only receive and vice versa.

Eg: Walkie Talkie , Internet Browsers

Data Transmission Mode

Full Duplex:



- Data can flow in both direction at the same time.
- it is **bidirectional**.
- Data can be sent in both directions simultaneously.
- We can send as well as we receive the data.

Eg: Telephone Network where both persons can talk and listen at the same time.

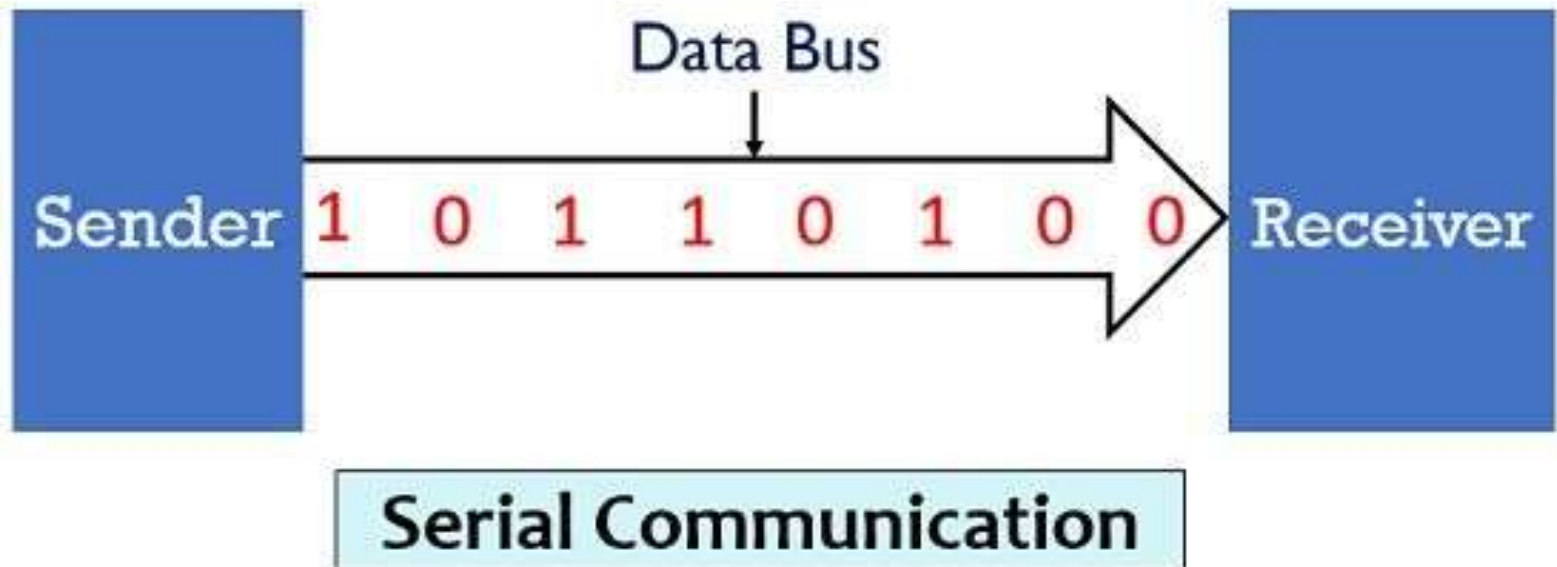
Data Transmission Methods

There are two methods used for transferring data between computers which are given below:

- **Serial Transmission**
- **Parallel Transmission**

Serial Transmission

- Data is sent one bit at a time.
- We need only one channel rather than n transmission lines to transmit data between two communicating devices.
- It is cheap but slow.



Serial Transmission

Advantages:

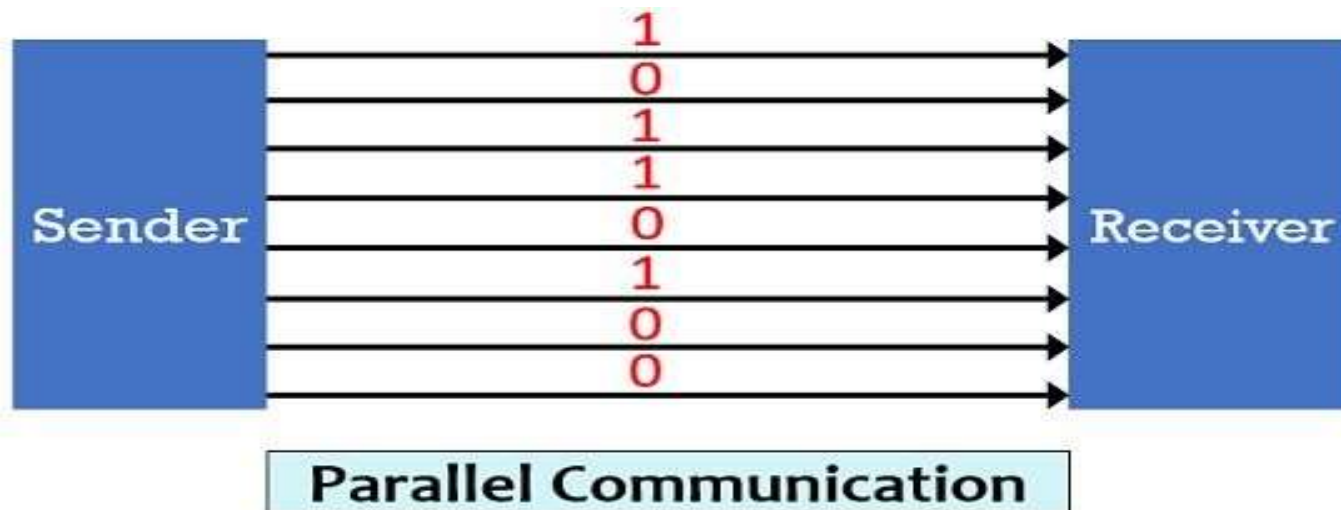
- This method is very reliable due to high data integrity.
- Best for long distance transmission.
- Cost effective.

Disadvantages:

- It is slow as one bit at a time is transmitted.
- As communication within devices is parallel, we require conversion devices at the interfaces.

Parallel Transmission

- Data is sent with multiple bits at a time, across multiple channels.
- We need multiple transmission lines to transmit data between two communicating devices.
- Its costly but fast.



Parallel Transmission

Advantages:

- Speed is high.

Disadvantages:

- Suitable for short distances only.
- Due to multiple channels lack of data integrity.
- Its costly.

Serial Vs. Parallel Transmission

Serial	Parallel
Single communication line is used for communication	Multiple lines are used
Cost efficient	Costly
One bit transferred at one clock pulse	8 bits transferred at one clock pulse
Used for long distance: Data transmission between two computers	Used for short distance Eg: data transmission between computer and a printer
Simple circuits	Complex circuits
More reliable	Less reliable due to crosstalk

Serial Transmission

- Based on Synchronization between two devices the serial data transmission method is classified into 2 types:

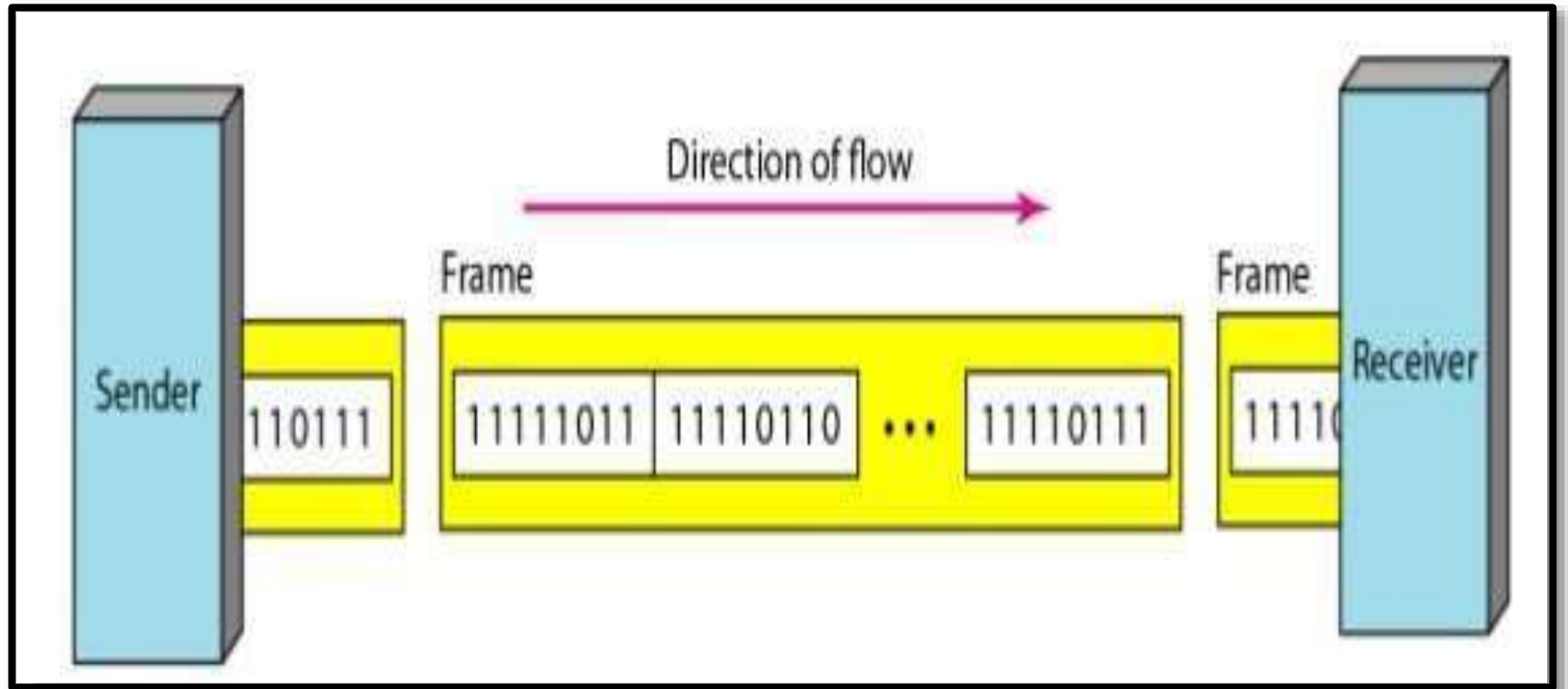
1. **Synchronous**
2. **Asynchronous**

Serial Transmission

Synchronous Transmission

- Sending bits one after the other **without start/stop bits or gaps.**
- Data flows in **form of frames.**
- For synchronization external clock is used.
- Data transfer rate is high.
- Costly.

Serial Transmission



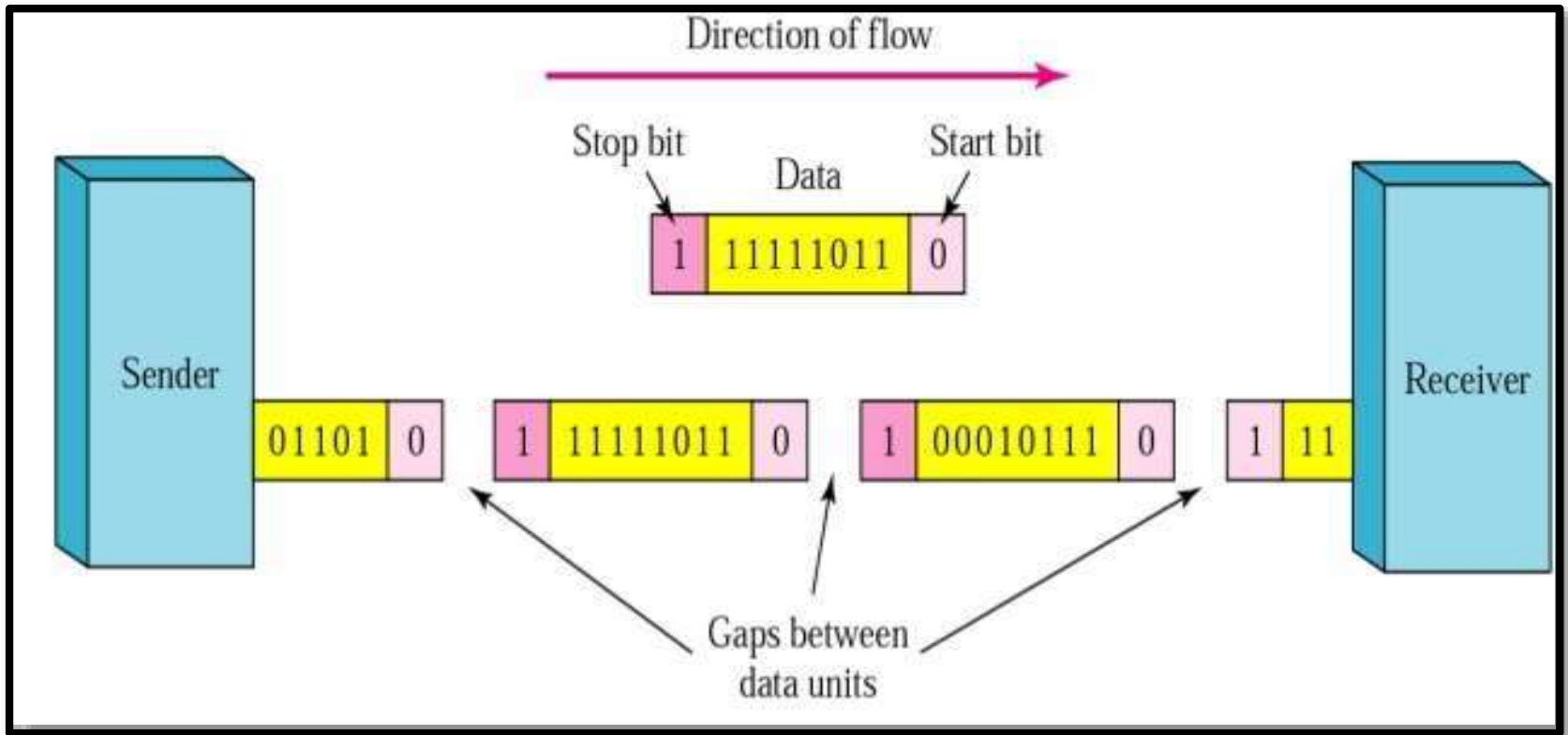
Synchronous Transmission

Serial Transmission

Asynchronous Transmission

- Sending bytes one after the other **with start/stop (parity) bits and gaps.**
- Data flows **one byte at a time.**
- For synchronization external clock is not used.
- Data transfer rate is slow because of parity bits and gaps.
- Cost effective.

Serial Transmission

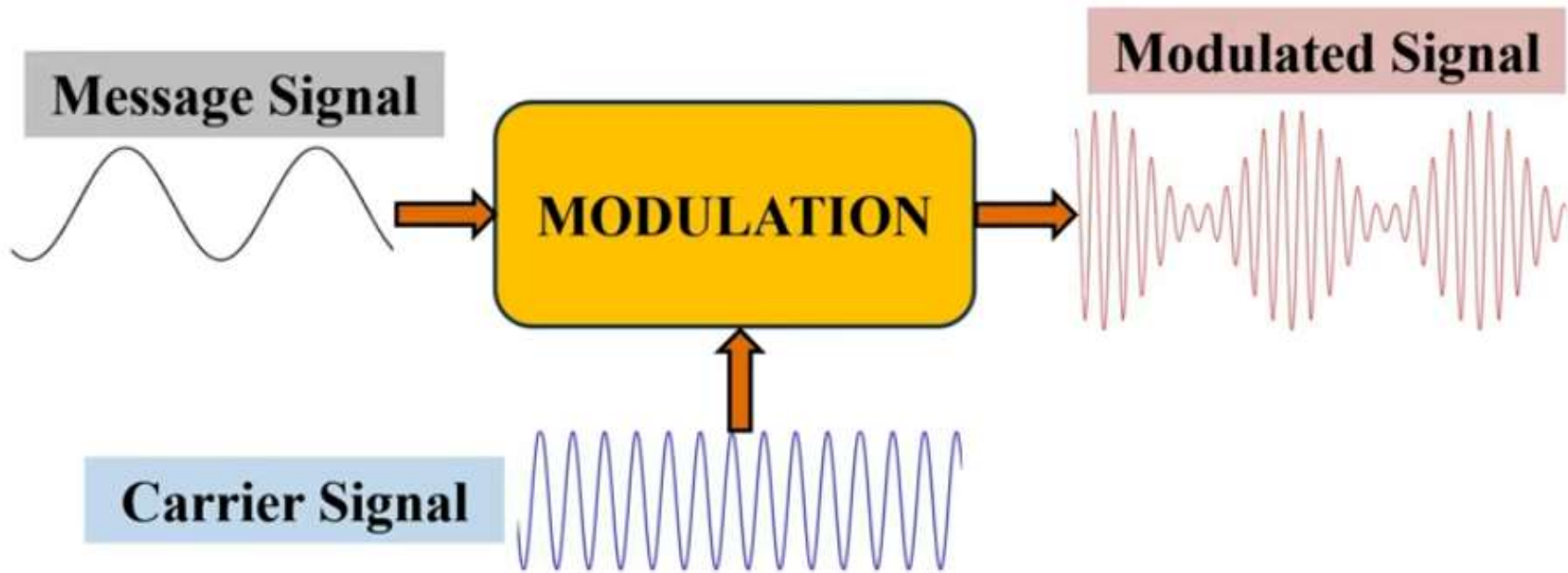


Asynchronous Transmission

Synchronous Vs. Asynchronous Transmission

Synchronous	Asynchronous
Data sent in form of frames	Data sent in form of bytes
Requires clock signal between the sender and receiver to inform the receiver about the new byte	Clock is not required as the data is sent and received with the help of parity bit attached which indicates the start of new byte
Fast	Slow
Costly	Economic
Efficient as no gaps	Less efficient due to gaps

What is Modulation?



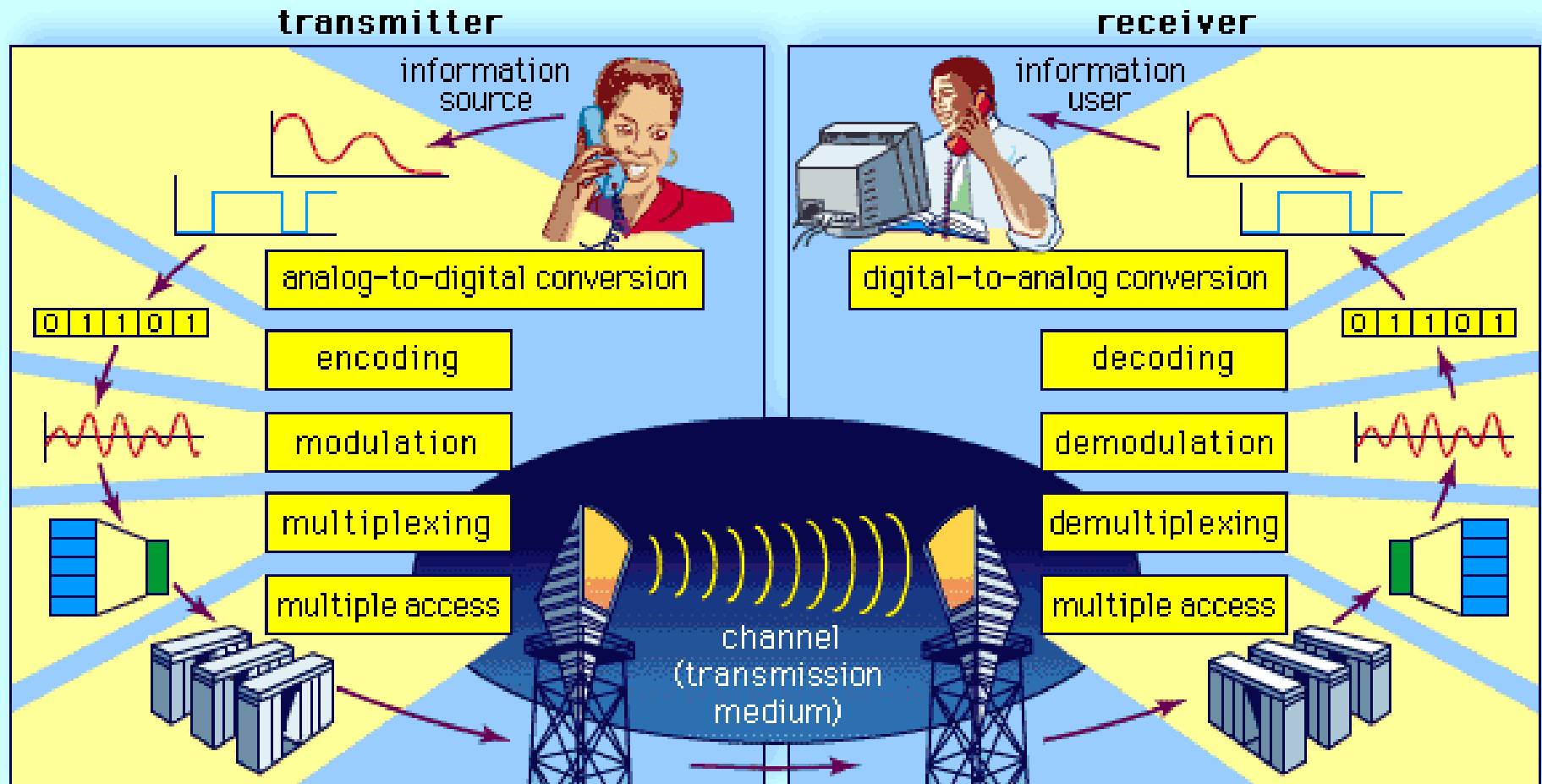
Modulation: It is a process of superimposing low signal into a high frequency carrier wave so that data can travel far distances.

What is Modulation?

Modulation:

- Modulation is a process of changing one or more characteristics of the carrier signal in proportion with the information signal. In modulation some parameter of the carrier wave(amplitude, frequency or phase) is varied in accordance with the information signal. This modulated signal is then transmitted by the transmitter.
- Carrier Waves are high frequency waves that helps in travelling long distances.

Where is it used?



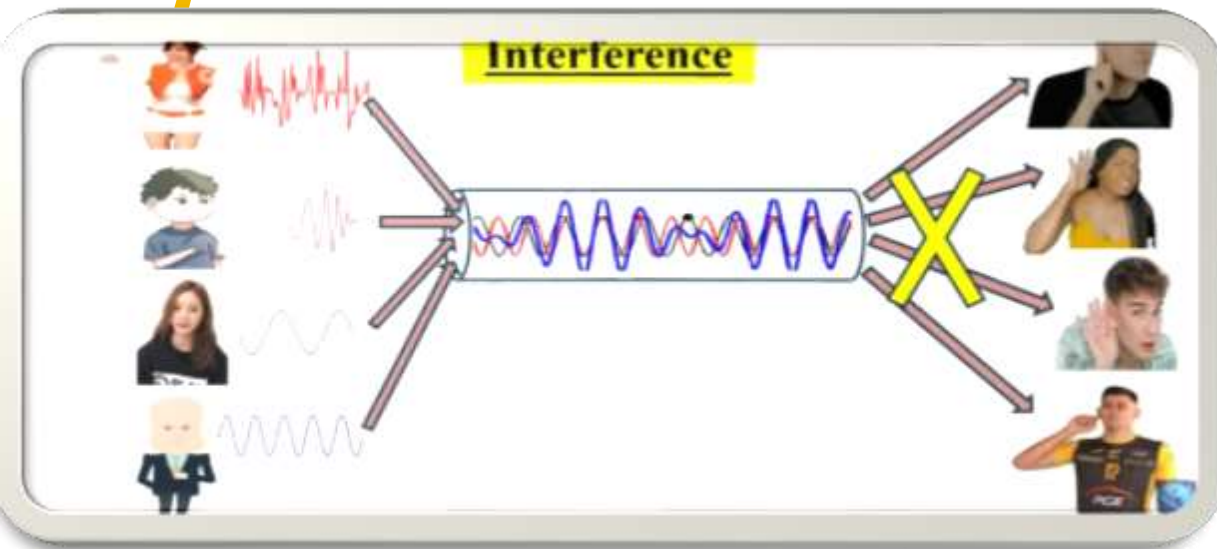
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They are used in telecommunication system. Eg: Radio System, Television, Mobile Network, etc.

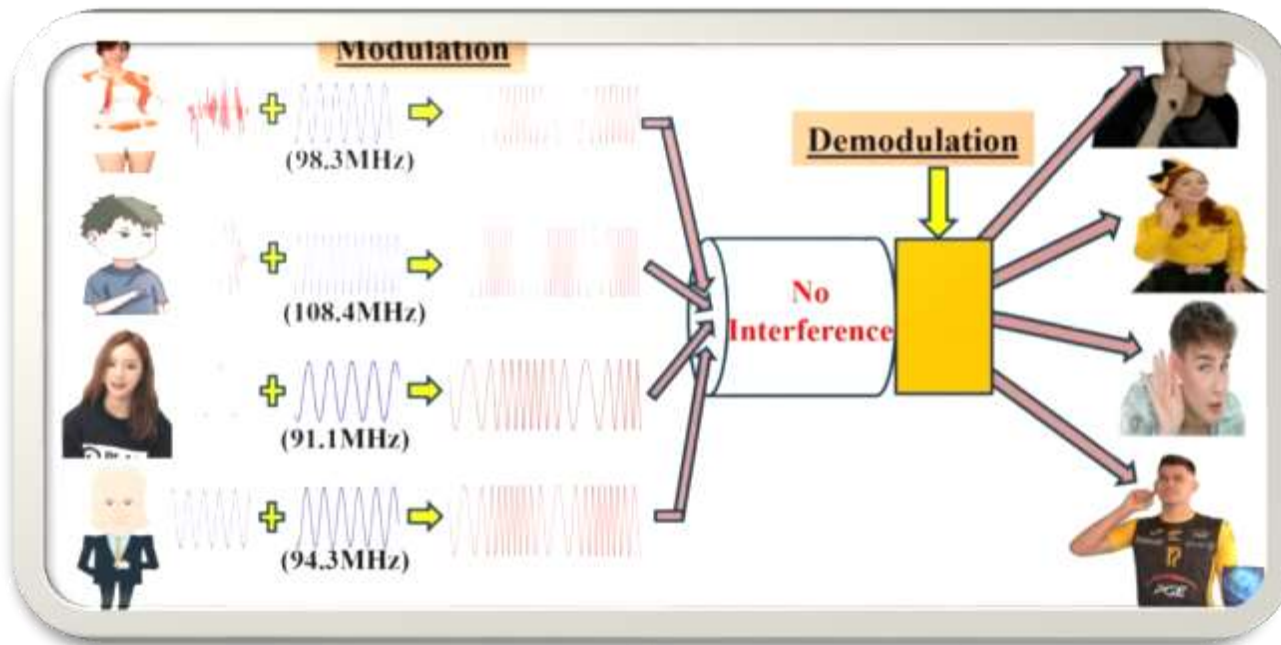
Why Modulation is used in Communication?

- To support multiplexing
- To decrease the size of antenna
- To allow Baseband signals (low frequency signals) to travel longer distances.
- To avoid interference of multiple signals and reduce noise in channel.
- The range of communication can be enhanced.

Why Modulation is used in Communication?

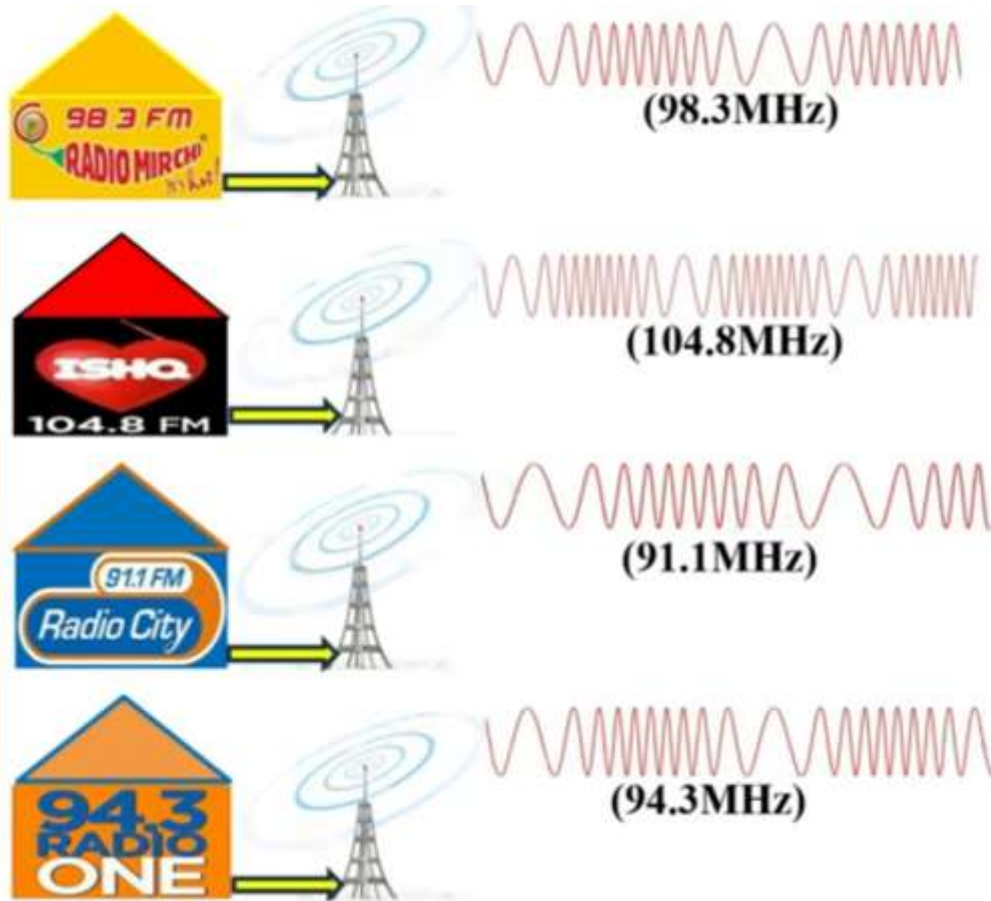


- To avoid interference of multiple signals and reduce noise in channel.



Why Modulation is used in Communication?

Modulation



Reduced Interference
Frequency Multiplexing



Demodulation

To allow multiplexing.

Why Modulation is used in Communication?

Size of the Antenna

Voice signal

350Hz to 3KHz



$$\text{Antenna size (L)} = \lambda / 2$$

$$F = 3 \text{ KHz}$$

$$C = 3 \times 10^8 \text{ m/s}$$

$$C = \lambda \times F$$

$$\lambda = \frac{C}{F}$$

$$\lambda = \frac{3 \times 10^8 \text{ m/s}}{3 \text{ KHz}} = \frac{3 \times 10^8}{3 \times 10^3} = 10^5 = 100 \text{ Km}$$

$$\text{Antenna size (L)} = \frac{100}{2} = \underline{\underline{50 \text{ Km}}}$$

- To decrease the size of antenna as size of antenna depends on wavelength of a signal.

Why Modulation is used in Communication?

Size of the Antenna



$$\text{Antenna size (L)} = \lambda / 2$$

$$F = 30 \text{ MHz}$$

$$C = 3 \times 10^8 \text{ m/s}$$

$$\lambda = \frac{3 \times 10^8}{30 \times 10^6} = 10 \text{ m}$$

$$\text{Antenna size (L)} = \frac{10}{2} = \underline{5 \text{ m}}$$

$$C = \lambda \times F$$

$$\lambda = \frac{C}{F}$$

- Higher frequency lower the wavelength so smaller the size of antenna.

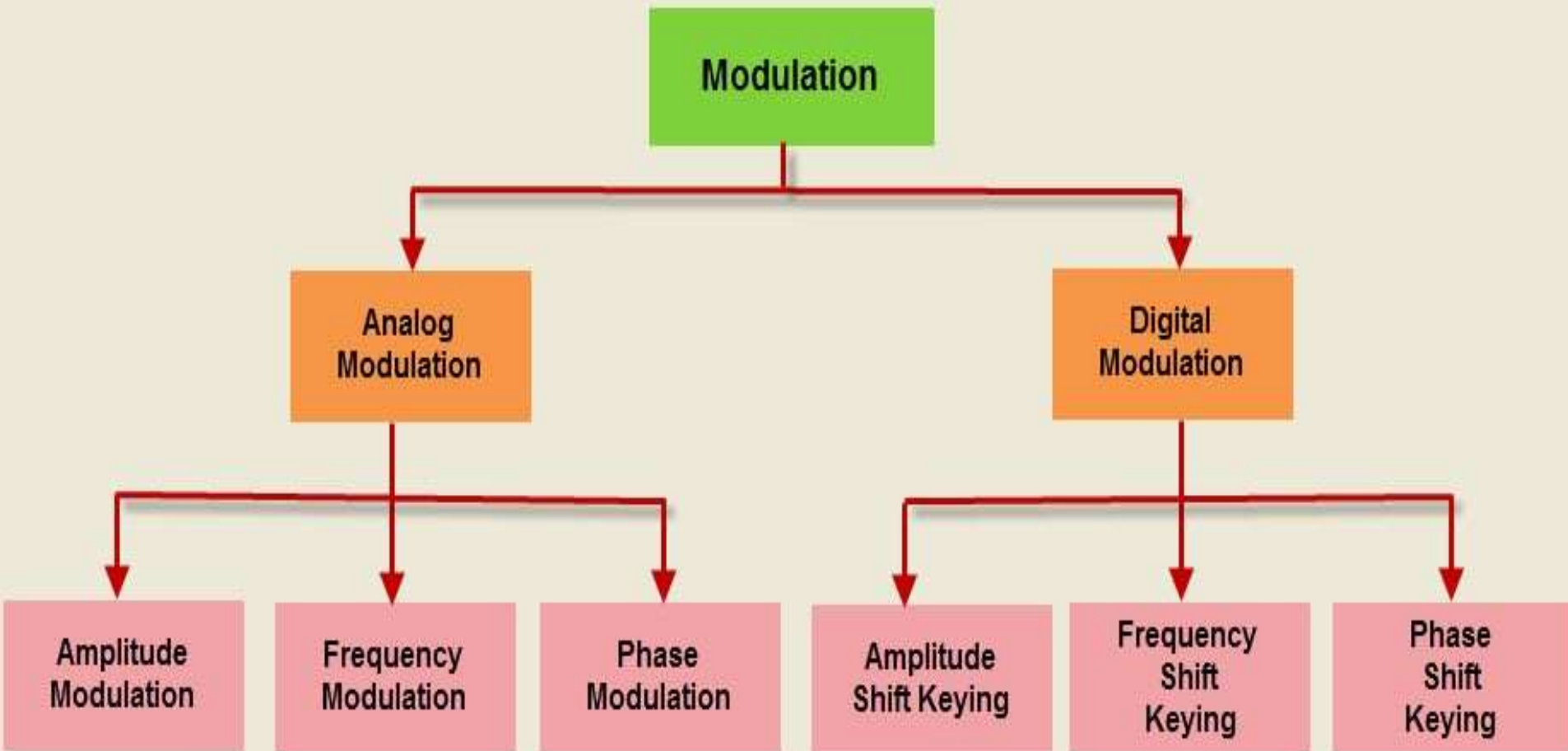
Components of Modulation system

- **Modulator** - It is circuit which performs modulation.
- **Modulating Signal** – Message, Information signal. It is the low frequency base signal that is to be transmitted.
- **Carrier Signal** – It is the high frequency signal used to carry information signal.
- **Modulated Signal** – It is a high frequency signal that is generated by imposing message on carrier signal.
- **Demodulator** – It is a circuit in receiver which performs demodulation.

Types of Modulation

- Depending on the type of communication system we have two types of Modulation Techniques
- **Analog Modulation**
- **Digital Modulation**

Types of Modulation



Analog Modulation

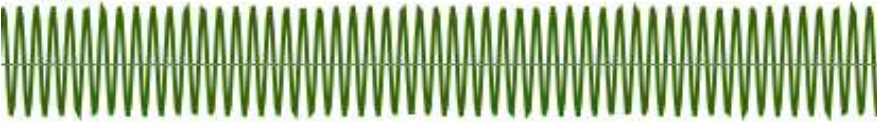
- In this modulation, message signal is Analog wave. A continuously varying sine wave is used as a carrier wave that modulates the analog message signal or data signal.
- **Types of Analog Modulation are**
 1. **AM** – Amplitude Modulation
 2. **FM** – Frequency Modulation
 3. **PM** – Phase Modulation

Analog Modulation

- **Amplitude Modulation** – In **AM**, the amplitude of the carrier wave is varied in proportion to the message signal, and the other factors like frequency and phase remain constant.
- **Frequency Modulation** – In **FM** the frequency of the carrier wave varies in proportion to the message or data signal while maintaining other parameters constant.
- **Phase Modulation** – In **PM** the carrier phase is varied in accordance with the data signal.

Analog Modulation

Carrier



Modulating Signal

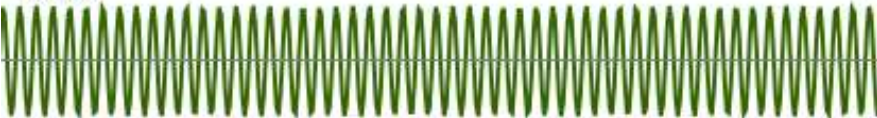


Modulated Signal



Amplitude Modulation

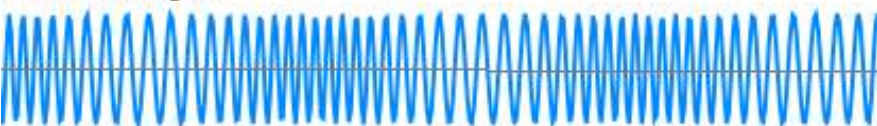
Carrier



Modulating Signal



Modulated Signal



Frequency Modulation

Carrier



Modulating Signal



Modulated Signal



Phase Modulation

Analog Modulation Drawback

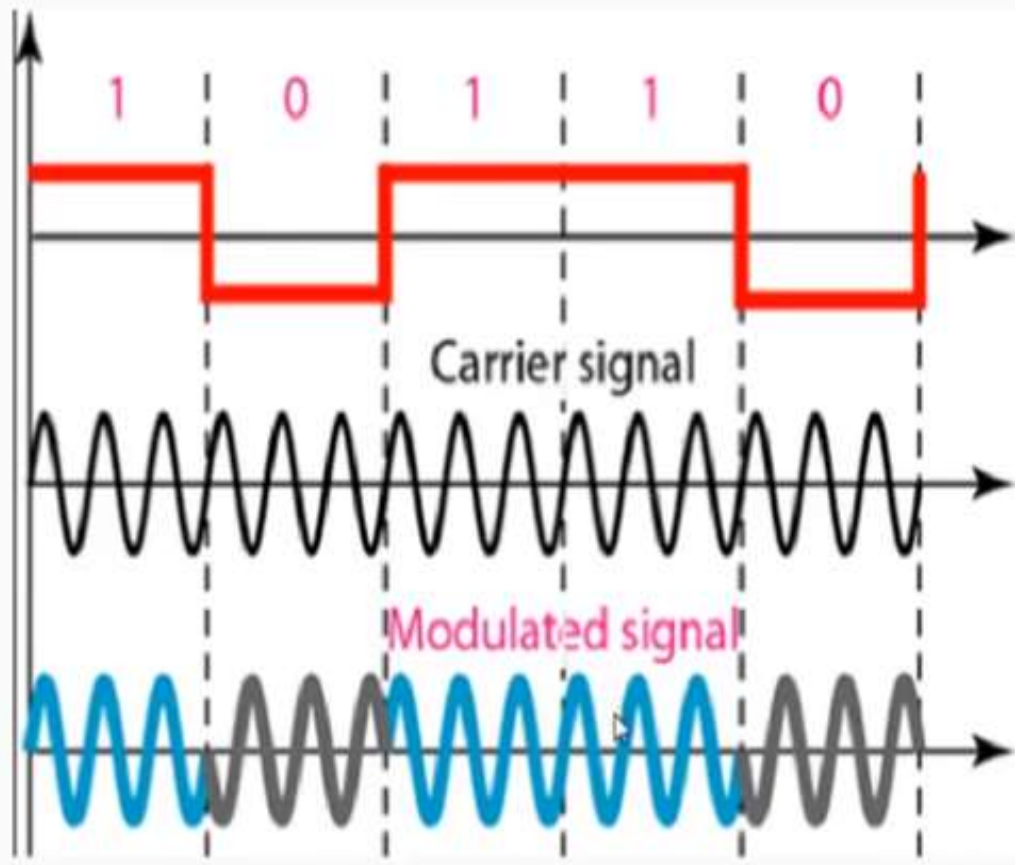
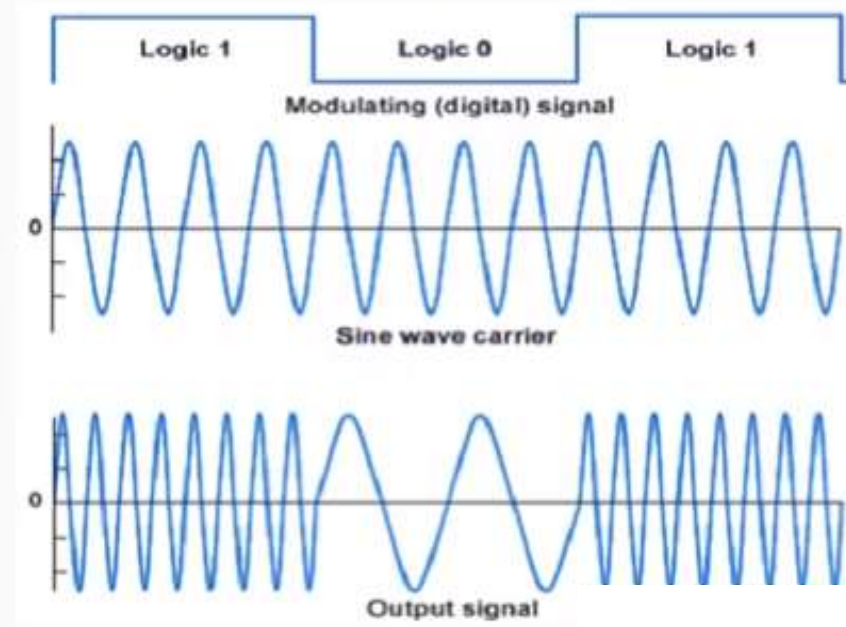
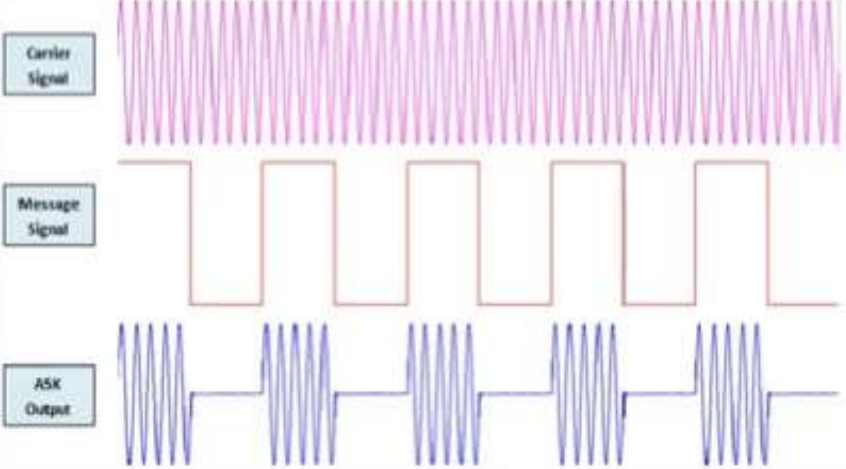
Analog modulation (AM, FM, and PM) is more sensitive to noise. If noise enters into a system, it persists and gets carried till the end receiver. Therefore, this drawback can be overcome by the digital modulation technique.

Digital Modulation

- For better quality and efficient communication, the digital modulation technique is employed.
- The main advantages of digital modulation over analog modulation include permissible power, available bandwidth, and high noise immunity.
- In digital modulation, a message signal is converted from analog to digital message and then modulated by using a carrier wave.

Digital Modulation

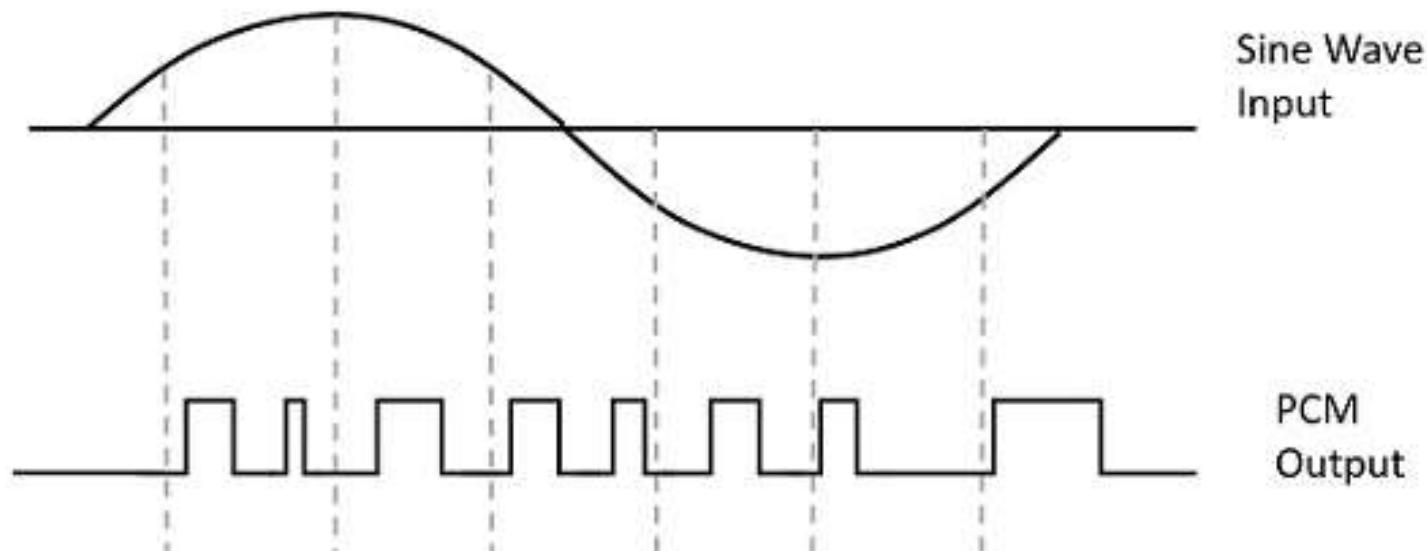
- **Amplitude Shift Keying** – ASK changes the amplitude of the carrier wave based on the baseband signal or message signal, which is in digital format. It is used for low-band requirements and is sensitive to noise.
- **Frequency Shift Keying** – In FSK, the frequency of the carrier wave is varied for each symbol in the digital data. It needs large bandwidths.
- **Phase Shift Keying** – The PSK changes the phase of the carrier for each symbol and it is less sensitive to noise.



Digital Modulation

Pulse Code Modulation (PCM)

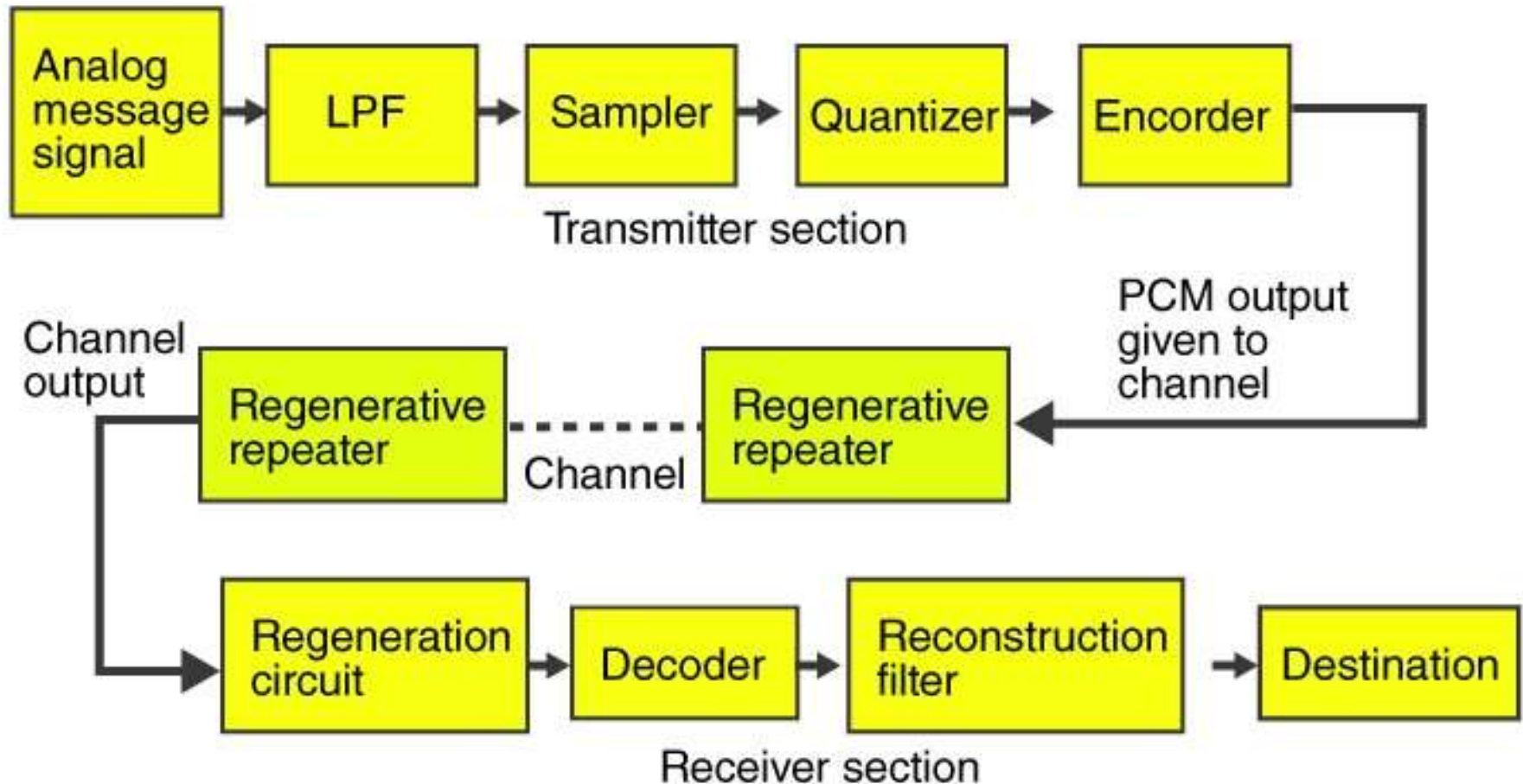
A signal is pulse code modulated to convert its analog information into a binary sequence, i.e., **1s** and **0s**. The output of a PCM will resemble a binary sequence. The following figure shows an example of PCM output with respect to instantaneous values of a given sine wave.



Pulse Code Modulation (PCM)

- Instead of a pulse train, PCM **produces a series of numbers or digits**, and hence this process is called as **digital**.
- Each one of these digits, though in binary code, represent the approximate amplitude of the signal sample at that instant.
- In Pulse Code Modulation, the message signal is represented by a sequence of coded pulses. This message signal is achieved by representing the signal in discrete form in both time and amplitude.

Pulse Code Modulation (PCM)

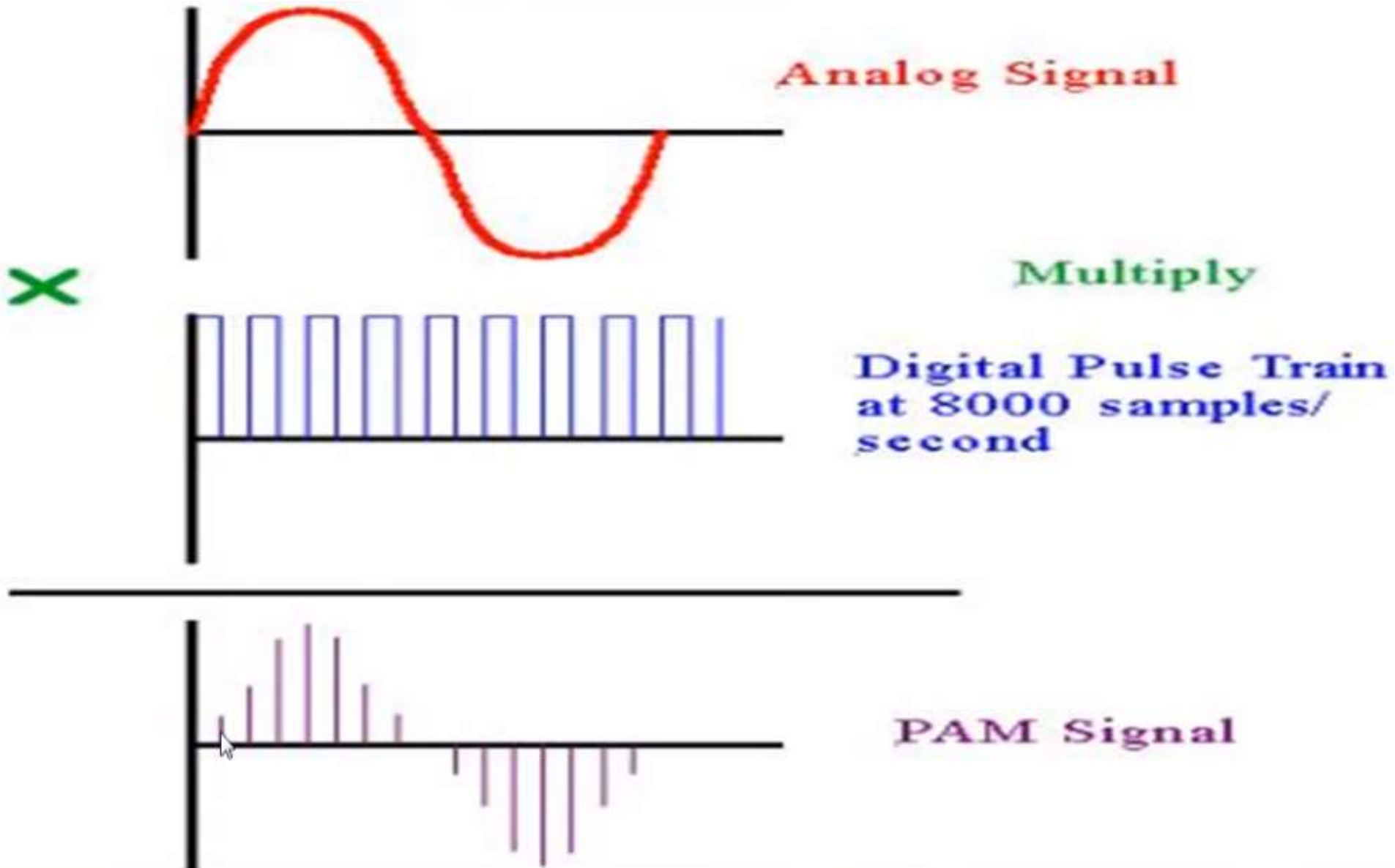


Pulse Code Modulation (PCM)

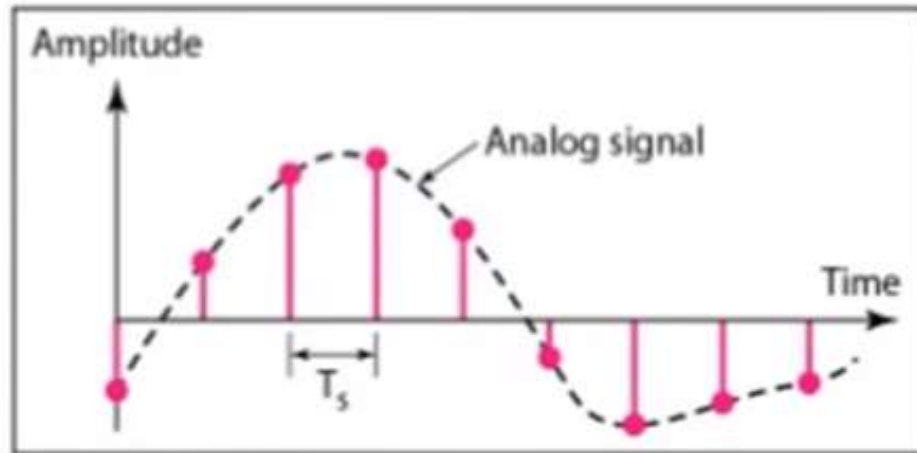
The Pulse Code Modulation process is done through the following steps:

- **Sampling** – To sample the continuous wave signal. It measures the amplitude of an Analog signal at regular interval on the basis of Nyquist Theorem.
- **Quantization** - It is the process to convert infinite value signal into a finite value. (Each sample is rounded off to a specific value.)
- **Encoding** – Finite values are assigned Binary codes
- The binary numbers are transmitted as series of pulses.

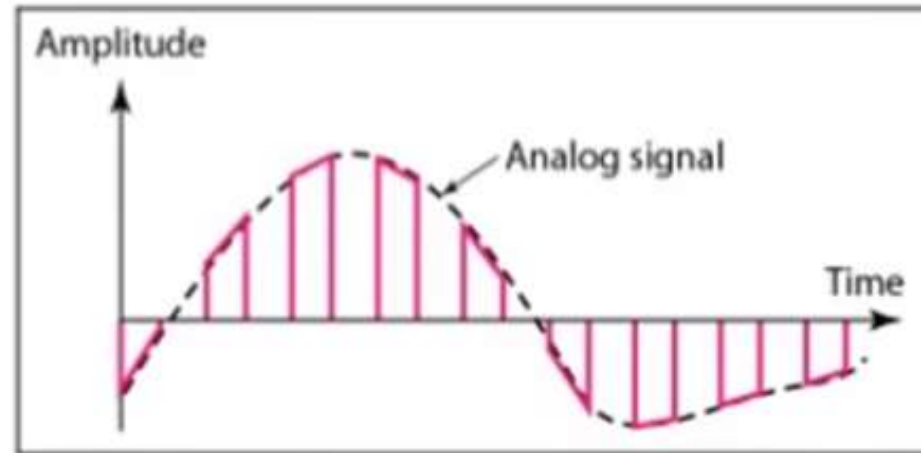
Pulse Code Modulation (PCM)



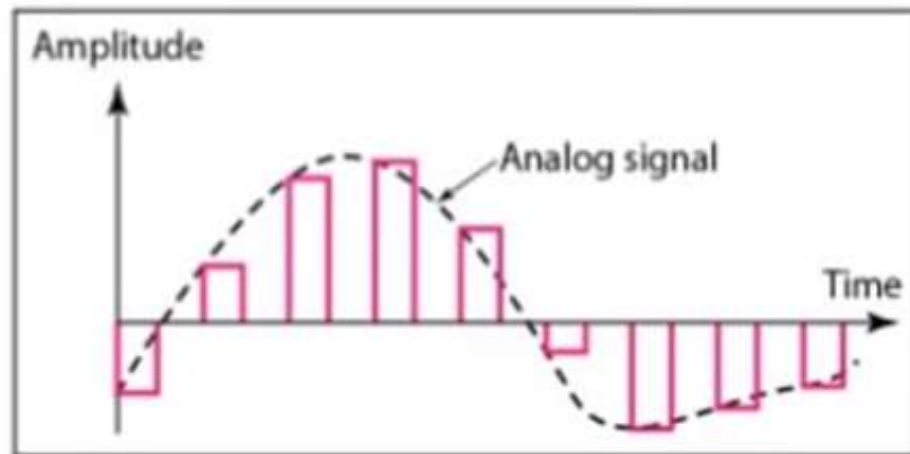
Pulse Code Modulation (PCM) - Sampling



a. Ideal sampling



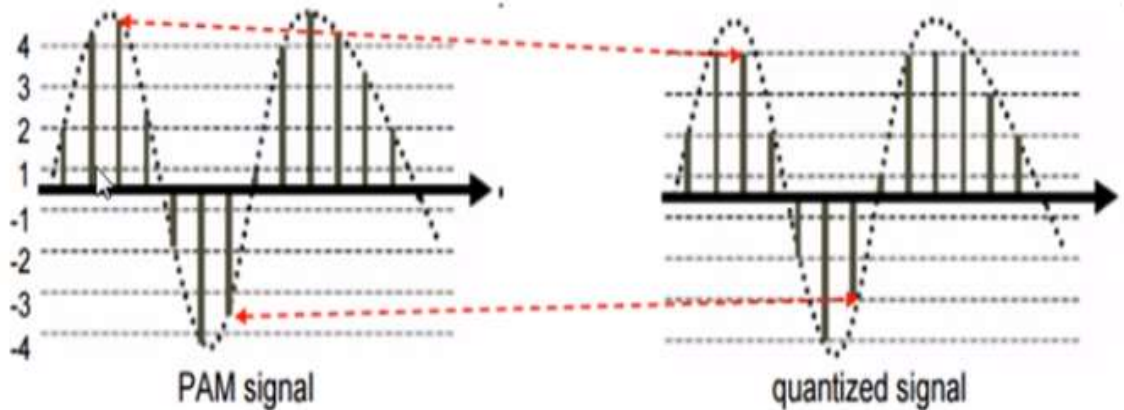
b. Natural sampling



c. Flat-top sampling

Pulse Code Modulation (PCM) - Quantization

It is the process to convert infinite value signal into a finite level.



Level	Code word
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

A/D output = n bits per sample (quantization level $M=2^n$)

Pulse Code Modulation (PCM)

- In the PCM process, it is possible to digitize all forms of analog data, including music, telemetry, voice, full-motion video. To obtain a pulse code modulated waveform from an analog waveform at the transmitter end and to convert the message signal into the binary form, a process known as quantization is used.
- At the receiver end of the pulse code circuit, demodulation takes place, and the signal is converted into pulses with the same quantum levels.

Pulse Code Modulation (PCM)

Low Pass Filter

This filter eliminates the high frequency components present in the input analog signal which is greater than the highest frequency of the message signal, to avoid aliasing of the message signal.

Sampler

This is the technique which helps to collect the sample data at instantaneous values of message signal, so as to reconstruct the original signal. As per the sampling theorem, the sampling rate is greater than the highest frequency component of the message signal.

Pulse Code Modulation (PCM)

Quantizer

Quantizing is a process of reducing the excessive bits and confining the data. The sampled output when given to Quantizer, reduces the redundant bits and compresses the value.

Encoder

The digitization of analog signal is done by the encoder. It designates each quantized level by a binary code. The sampling done here is the sample-and-hold process. These three sections LPF, Sampler and Quantizer will act as an analog to digital converter. Encoding minimizes the bandwidth used.

Pulse Code Modulation (PCM)

Regenerative Repeater

This section increases the signal strength. The output of the channel also has one regenerative repeater circuit, to compensate the signal loss and reconstruct the signal, and also to increase its strength.

Decoder

The decoder circuit decodes the pulse coded waveform to reproduce the original signal. This circuit acts as the demodulator.

Pulse Code Modulation (PCM)

Reconstruction Filter

After the digital-to-analog conversion is done by the regenerative circuit and the decoder, a low-pass filter is employed, called as the reconstruction filter to get back the original signal.

Hence, the Pulse Code Modulator circuit digitizes the given analog signal, codes it and samples it, and then transmits it in an analog form. This whole process is repeated in a reverse pattern to obtain the original signal.

PCM- Advantages Disadvantages

Advantages

- PCM is used in long-distance communication.
- The efficiency of the transmitter in PCM is high.
- Higher noise immunity is seen.
- Efficient method.

Disadvantages

- The bandwidth requirement is high.
- PCM is a complex process, since it involves encoding, decoding and quantization of the circuit.

PCM- Applications

- It is used in telephony and compact discs.
- Pulse Code Modulation is used in satellite transmission systems and space communications.